



Malawi

Malaria Indicator Survey

2021

Ministry of Health
National Malaria Control Programme
Lilongwe, Malawi

February 2022

The 2021 Malawi Malaria Indicator Survey (2021 MMIS) was implemented by the Malawi Ministry of Health through the National Malaria Control Programme (NMCP). Financial support for the survey was provided by the Global Fund (GF). The government of Malawi provided financial assistance in terms of in form of personnel, office space, and logistical support. National Statistic Office (NSO) provided technical assistance as it provides technical assistance in the implementation of population and health surveys in the country.

Additional information about the 2021 MMIS may be obtained from the National Malaria Control Programme, P/Bag 65, Mtunthama Drive, Lilongwe, Malawi.

Cover photos:

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PREFACE

Malaria is a major health problem in Malawi where an estimated 4 million cases occur each year. Children under age 5 and pregnant women are most likely to have severe illness. Ministry of Health, in collaboration with partners, developed the Malawi Health Sector Strategic Plan 2017-2022, which articulates the priorities for health sector development in the past 5 years and prioritizes malaria. In line with that emphasis, the National Malaria Control Program has just finishing the implementation of the developed National Malaria Strategic Plan 2017–2022 with the goal of scaling up malaria interventions to reduce morbidity and mortality by 50% in 2022. Progress would continue towards attainment of the national vision of a malaria-free Malawi.

We strive for progress in achieving prompt, effective malaria treatment. We hope to improve access to early intervention and treatment by expanding village clinic services, using insecticide-treated nets, spraying inside residences, managing the environment, encouraging changes in social behaviour and communication, and preventing malaria in pregnancy. We have set for ourselves high targets for these interventions, and we are confident that we will achieve our strategic goals of halving the incidence of malaria and deaths, as well as reducing the prevalence of malaria and malaria-related anaemia.

Surveys such as the current Malaria Indicator Survey (MIS) are essential measures of progress towards these goals. Without measurement, we can only guess about progress. The 2021 Malawi Malaria Indicator Survey (MMIS) is the country's fifth nationally representative assessment of the coverage attained by key malaria interventions. Interventions are reported in combination with measures of malaria-related burden and anaemia prevalence testing among children under age 5.

Overall, there has been considerable progress in scaling up interventions and controlling malaria. We noted a decline in malaria prevalence from 24% in 2017 to 10.5% in 2021. Results of the 2021 MMIS also show improvement on use of intermittent preventive treatment during pregnancy (IPTp) by pregnant women age 15-49. Coverage has increased from 77% for two or more doses in 2017 to 81% in 2021. The percentage of women who took three or more doses of SP/Fansidar for prevention of malaria in pregnancy increased from 43% in 2017 to 56% in 2021. In addition, numbers of children receiving a parasitological test and artemisinin-based combination therapy continue to increase. Despite all these achievements, insecticide-treated net (ITN) ownership has decreased from 82% in 2017 to 55% in 2021 because the 2021 MIS occurred 3 years after the 2018 mass LLIN distribution campaign unlike the 2017 MIS which occurred approximately one year after a partial mass LLIN distribution campaign.

These results represent the combined work of numerous partners contributing to the overall scale-up of malaria interventions. I would like to request that all partners make use of the information presented in this report as they implement projects to surmount the challenges depicted here.

Finally, I would like to thank the National Malaria Control Programme for taking a leading role in this survey and thank all of those who travelled to various areas of Malawi, including the most remote parts of the country, to collect data. Most important, I thank the survey respondents for their contributions to this survey. Together, we can make Malawi free of malaria.

Dr. Charles Mwansambo
Secretary for Health
Ministry of Health

ACRONYMS AND ABBREVIATIONS

ACT	artemisinin-based combination therapy
CAPI	computer-assisted personal interviewing
CCAP	Church of Central Africa Presbyterian
CHAM	Christian Health Association of Malawi
CSPro	Census and Survey Processing System
DHS	Demographic and Health Survey
EA	enumeration area
IPTp	intermittent preventive treatment (of malaria) in pregnancy
ITN	insecticide-treated net
LA	lumefantrine-artemether
MERG	monitoring and evaluation reference group
MMIS	Malawi malaria indicator survey
MIS	malaria indicator survey
MoH	Ministry of Health
NMCP	national malaria control programme
NSO	national statistical office
PMI	US President's Malaria Initiative
RBM	Roll Back Malaria
RDT	rapid diagnostic test
SP	sulphadoxine-pyrimethamine
USAID	United States Agency for International Development
WHO	World Health Organization

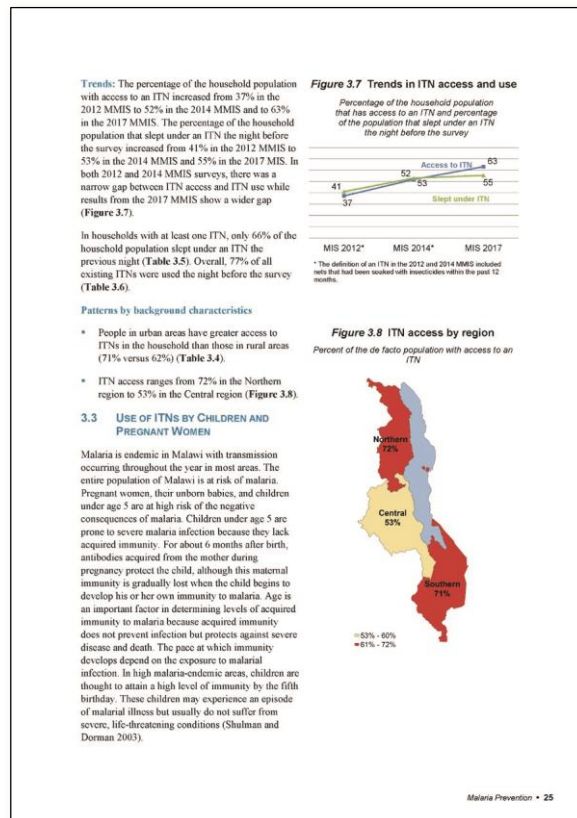
Acronyms and Abbreviations •

READING AND UNDERSTANDING TABLES FROM THE 2017 MALAWI MALARIA INDICATOR SURVEY (MMIS)

The 2021 Malawi Malaria Indicator Survey (MMIS) report is very similar in content to the 2017 MMIS is presented in a similar format. The style features more figures to highlight trends, subnational patterns, and background characteristics. Large colourful maps display data by regions within Malawi. The text has been simplified to highlight key points in bullets and to clearly identify indicator definitions in boxes.

Although the text and figures featured in each chapter highlight some of the most important findings, not every finding can be discussed or displayed graphically. For this reason, 2021 MMIS data users should be comfortable reading and interpreting data tables.

The following pages provide an introduction to the organization of the 2021 MMIS tables, the presentation of background characteristics, and a brief summary of sampling procedures used and understanding denominators. In addition, this section provides some exercises for users as they practice their new skills in interpreting 2021 MMIS tables.



Example 1: Low Haemoglobin

A measure taken from all eligible respondents

Table 4.5 Haemoglobin <8.0 g/dl in children			1
Percentage of children age 6-59 months with haemoglobin lower than 8.0 g/dl, by background characteristics, Malawi MIS 2017			
Background characteristic	Haemoglobin < 8.0 g/dl	Number of children	2
Age in months			
6-8	10.0	141	
9-11	8.0	120	
12-17	8.1	278	
18-23	10.8	227	
24-35	3.1	517	
36-47	4.4	528	
48-59	2.4	674	
Sex			
Male	4.1	1,246	
Female	6.1	1,238	
Mother's interview status			
Interviewed	5.1	2,183	
Not interviewed	4.7	302	
Residence			
Urban	4.3	339	
Rural	5.2	2,145	5
Region			
Northern	3.5	268	
Central	6.1	1,065	
Southern	4.5	1,152	
Mother's education¹			
No education	5.0	328	
Primary	5.6	1,479	
Secondary	3.9	347	
More than secondary	0.0	28	
Wealth quintile			
Lowest	5.0	588	
Second	4.9	579	
Middle	7.7	467	
Fourth	4.9	443	
Highest	2.6	408	
Total	5.1	2,485	4
Note: Table is based on children who stayed in the household the night before the interview. Prevalence of anaemia is based on haemoglobin levels and is adjusted for altitude using CDC formulas (CDC 1998). Haemoglobin is measured in grams per decilitre (g/dl).			
¹ Excludes children whose mothers are not interviewed.			

Step 1: Read the title and subtitle. They tell you the topic and the specific population group being described. In this case, the table is about anaemia in children (haemoglobin <8.0 g/dl). Haemoglobin levels were measured for all eligible children age 6-59 months whose parents or guardians gave their consent.

Step 2: Scan the column headings—highlighted in green in Example 1. They describe how the information is categorized. In this table, the first column of data shows children who have moderate-to-severe anaemia, or haemoglobin <8.0 g/dl. The second column lists the number of children age 6-59 months who were tested for low haemoglobin in the survey.

Step 3: Scan the row headings—the first vertical column highlighted in blue in Example 1. These show the different ways the data are divided into categories based on population characteristics. In this case, the table presents prevalence of low haemoglobin by age, sex, mother's interview status, urban-rural residence, region, mother's educational level, and wealth quintile.

Step 4: Look at the row at the bottom of the table highlighted in red. These percentages represent the totals of children age 6-59 months with low haemoglobin. In this case, 5.1%* of children age 6-59 months had haemoglobin <8.0 g/dl.

Step 5: To find out what percentage of children age 6-59 months in rural areas had low haemoglobin, draw two imaginary lines, as shown on the table. This shows that 5.2% of children age 6-59 months in rural areas had haemoglobin <8.0 g/dl.

Step 6: By looking at patterns by background characteristics, we can see how low haemoglobin varies across Malawi. Resources are often limited; knowing how malaria-related anaemia varies among different groups can help programme planners and policy makers determine how to most effectively use resources.

**For the purpose of this tutorial, data are presented exactly as they appear in the table including decimal places. However, the text in the remainder of this report rounds data to the nearest whole percentage point.*

Practice: Use the table in Example 1 to answer the following questions about low haemoglobin:

- Is low haemoglobin more common among boys or girls?
- Is there a clear pattern of low haemoglobin by age?
- What are the lowest and highest percentages (range) of low haemoglobin by region?
- Is there a clear pattern of low haemoglobin by mother's education level?

Answers:

a) Low haemoglobin is slightly higher among girls (6.1%) than among boys (4.1%).

b) Low haemoglobin is highest among children age 18 months (10.8%) and then decreases as children get older.

c) Low haemoglobin is least common in Northern Region (3.5%) and most common in Central Region (6.1%).

d) There is no clear pattern by mother's level of education; children whose mothers have primary education have the greatest levels of low haemoglobin (5.6%).

Example 2: Use of Mosquito Nets by Pregnant Women

A Question Asked of a Subgroup of Survey Respondents

Table 3.8 Use of mosquito nets by pregnant women					
Percentage of pregnant women age 15-49 who, the night before the survey, slept under a mosquito net (treated or untreated) and under an insecticide-treated net (ITN); and among pregnant women age 15-49 in households with at least one ITN, percentage who slept under an ITN the night before the survey, according to background characteristics, Malawi MIS 2017					
Background characteristic	Among pregnant women age 15-49 in all households			Among pregnant women age 15-49 in households with at least one ITN ¹	
	Percentage who slept under any mosquito net last of night	Percentage who slept under an ITN ¹ last night	Number	Percentage who slept under an ITN ¹ last night	Number of pregnant women
Residence					
Urban	67.9	61.5	36	71.1	31
Rural	63.9	62.7	248	73.3	212
Region					
Northern	59.9	54.7	27		
Central				63.0	24
Southern					
Education					
No education					
Primary					
Secondary					
More than secondary					
Wealth quintile					
Lowest	59.6	57.5	129	74.5	99
Second					
Middle					
Fourth	70.2	69.3	128	73.7	120
Highest					
Total					
	*	*	26	*	23
	64.6	63.5	201	72.9	175
	64.5	58.4	52	74.8	41
	*	*	5	*	5
	(53.2)	(53.2)	67	(71.5)	50
	(67.2)	(67.2)	64	(73.5)	59
	(74.0)	(71.2)	59	(78.4)	54
	55.8	54.7	48	(61.8)	42
	73.3	66.3	46	38	79.1
					243
	64.4	62.5	284	73.0	
Note: Table is based on women who stayed in the household the night before the interview. Figures in parentheses are based on 25-49 unweighted cases. An asterisk indicates that a figure is based on fewer than 25 unweighted cases and has been suppressed.					
¹ (ITN) is a factory-treated net that does not require any further treatment. In the 2012 Malawi MIS and 2014 Malawi MIS long (LLIN) is a long-lasting insecticidal net (LLIN).					

Step 1: Read the title and subtitle. In this case, the table is about two separate groups of pregnant women: all pregnant women age 15-49 in all households (a) and pregnant women age 15-49 in households with at least one insecticide-treated net (ITN) (b).

Step 2: Identify the two panels. First, identify the columns that refer to all pregnant women age 15-49 in all households (a), and then isolate the columns that refer only to pregnant women age 15-49 in households with at least one ITN (b).

Step 3: Look at the number of women included in this table. How many pregnant women age 15-49 in all households were interviewed? It's 284. Now look at the second panel. How many pregnant women age 15-49 in households with at least one ITN were interviewed? It's 243.

Step 4: Only 284 pregnant women age 15-49 in all households and 243 pregnant women in households with at least one ITN were interviewed in the 2017 MMIS. Once these pregnant women are further divided into the background characteristic categories, there may be too few cases for the percentages to be reliable.

- What percentage of pregnant women age 15-49 in all households in the lowest wealth quintile slept under an ITN the night before the survey? 53.2%. This percentage is in parentheses because there are between 25 and 49 pregnant women (unweighted) in this category. Readers should use this number with caution—it may not be reliable. (For more information on weighted and unweighted numbers, see Example 3.)
- What percentage of pregnant women age 15-49 with no education in households with at least one ITN slept under an ITN the night before the survey? There is no number in this cell—only an asterisk. This is because fewer than 25 pregnant women with no education in households with at least one ITN were interviewed in the survey. Results for this group are not reported. The subgroup is too small, and therefore the data are not reliable.

Note: When parentheses or asterisks are used in a table, the explanation will be noted under the table. If there are no parentheses or asterisks in a table, you can proceed with confidence that enough cases were included in all categories that the data are reliable.

Example 3: Understanding Sampling Weights in 2021 MMIS Tables

A sample is a group of people who have been selected for a survey. In the 2021 MMIS, the sample is designed to represent the national population age 15-49. In addition to national data, most countries want to collect and report data on smaller geographical or administrative areas. However, doing so requires a minimum sample size per area. For the 2021 MMIS, the survey sample is representative at the national and regional levels, and for urban and rural areas.

To generate statistics that are representative of the country as a whole and the three

regions, the number of women surveyed in each region should contribute to the size of the total (national) sample in proportion to size of the region. However, if some regions have small populations, then a sample allocated in proportion to each region's population may not include sufficient women from each region for analysis. To solve this problem, regions with small populations are oversampled. For example, let's say that you have enough money to interview 3,860 women and want to produce results that are representative of Malawi as a whole and its regions (as in Table 2.8). However, the total population of Malawi is not evenly distributed among the regions: some regions, such as Southern Region, are heavily populated while others, such as Northern Region are not. Thus, Northern Region must be oversampled.

A sampling statistician determines how many women should be interviewed in each region in order to get reliable statistics. The **blue column (1)** in the table at the right shows the actual number of women interviewed in each region. Within the regions, the number of women interviewed ranges from 1,232 in Southern Region to 1,315 in Northern Region. The number of interviews is sufficient to get reliable results in each region.

With this distribution of interviews, some regions are overrepresented and some regions are underrepresented. For example, the population in Southern Region is 45% of the population in Malawi, while Northern Region's population contributes only 12%. But as the blue column shows, the number of women interviewed in Southern Region accounts for only about 32% of the total sample of women interviewed ($1,232/3,860$) and the number of women interviewed in Northern Region accounts for 34% of the total sample of women interviewed ($1,315/3,860$). This unweighted distribution of women does not accurately represent the population.

In order to get statistics that are representative of Malawi, the distribution of the women in the sample needs to be weighted (or mathematically adjusted) such that it resembles the true distribution in the country. Women from a small region, such as Northern Region, should only contribute a small amount to the national total. Women from a large region, like Southern Region, should contribute much more. Therefore, DHS statisticians mathematically calculate a "weight" which is used to adjust the number of women from each region so that each region's contribution to the total is proportional to the actual population of the region. The numbers in the **purple column (2)** represent the "weighted" values. The weighted values can be smaller or larger than the unweighted values at regional level. The total national sample size of 3,860 women has not changed after weighting, but the distribution of the women in the regions has been changed to represent their contribution to the total population size.

How do statisticians weight each category? They take into account the probability that a woman was selected in the sample. If you were to compare the **green column (3)** to the actual population distribution of Malawi, you would see that women in each region are contributing to the total sample with the same weight that they contribute to the population of the country. The weighted number of women in the survey now accurately

Table 2.8 Background characteristics of respondents

Percent distribution of women age 15-49, by selected background characteristics, Malawi MIS 2021

Background characteristic	Women		
	Weighted percent	Weighted number	Unweighted number
Region	3	2	1
Northern	12.0	465	1,315
Central	42.9	1,657	1,313
Southern	45.0	1,738	1,232
Total 15-49	100.0	3,860	3,860

represents the proportion of women who live in Northern Region and the proportion of women who live in Southern Region.

With sampling and weighting, it is possible to interview enough women to provide reliable statistics at national and regional levels. In general, only the weighted numbers are shown in each of the MMIS tables, so don't be surprised if these numbers seem low: they may actually represent a larger number of women interviewed.

MALAWI



Map of Malawi

The 2021 Malawi Malaria Indicator Survey (MMIS) was implemented by the Ministry of Health through the National Malaria Control Program (NMCP) with funding from the Global Fund (GF). Data collection took place from 12 April to 21 June, 2021. NSO provided technical assistance as it provides technical assistance in the implementation of population and health surveys in the country.

1.1 SURVEY OBJECTIVES

The 2021 MMIS, a comprehensive, nationally-representative household survey, was designed in accord with the Roll Back Malaria Monitoring and Evaluation Working Group (RBM-MERG) guidelines. The primary objective of the 2021 MMIS project was to provide up-to-date estimates of basic demographic and health indicators related to malaria. Specifically, the 2021 MMIS collected information on mosquito nets, intermittent preventive treatment of malaria in pregnant women (*IPTp*), and care seeking behaviour and treatment of fever in children. Young children were also tested for anaemia and for malaria infection. Knowledge of malaria was assessed among interviewed women. The information collected through the 2021 MMIS is intended to assist policy makers and program managers in evaluating and designing programs and strategies for improving the health of the country's population.

1.2 SAMPLE DESIGN

The 2021 MMIS followed a two-stage sample design and allows estimates of key malaria indicators for the country as a whole, for urban and rural areas separately, and for each of the 3 administrative regions in Malawi: Northern, Central, and Southern. The *first stage* of sampling involved selecting sample points (clusters/enumeration areas) from the sampling frame. There were 18, 483 enumeration areas (EAs) delineated for the 2018 Population and Housing Census that were used as the sampling frame. A total of 150 clusters were selected, with probability proportional to size, from the EAs covered in the 2018 Population and Housing Census. Of these clusters, 60 were in urban areas and 90 in rural areas. Urban areas were oversampled within regions to produce robust estimates for each area or domain. A household listing operation was undertaken in all selected EAs between February and March 2021, and households to be included in the survey were randomly selected from these lists in the second stage. The *second stage* of sampling involved equal probability systematic selection of households from the updated listed households. Twenty-five households were selected from each EA, for a total sample size of 3,750 households that were included in the interviews. Because of the approximately equal sample sizes in each region, the sample is not self-weighting at the national level. Results shown in this report have been weighted to account for the complex sample design. See Appendix A for additional details on the sampling procedures. All women age 15-49 who were either permanent residents of the selected households or visitors who stayed in the household the night before the survey were eligible to be interviewed. With the parent's or guardian's consent, children age 6-59 months were tested for anaemia and for malaria infection.

1.3 QUESTIONNAIRES

Data was primarily collected using three types of questionnaires: the Household Questionnaire, the Woman's Questionnaire, and the Biomarker Questionnaire. Core questionnaires available from the RBMMERG were adapted to reflect the population and health issues relevant to Malawi. The modifications were determined at a series of meetings with various stakeholders from the National Malaria Control Programme (NMCP) and other government ministries and agencies, nongovernmental organisations, and international donors. The questionnaires in English and two local Malawian languages (Chichewa and Chitumbuka) were programmed onto tablet computers, enabling use of computer-assisted personal interviewing (CAPI) for the survey.

The Household Questionnaire was used to list all the usual members and visitors to selected households. Basic information was collected on the characteristics of each person listed in the household, including his or her age, sex, and relationship to the head of the household. The data on the age and sex of household members, obtained from the Household Questionnaire, were used to identify women eligible for an individual interview and children age 6-59 months eligible for anaemia and malaria testing. Additionally, the Household Questionnaire captured information on characteristics of the household's dwelling unit, such as the source of water, type of toilet facilities, materials used for flooring, ownership of various durable goods, and ownership and use of mosquito nets.

The Woman's Questionnaire was used to collect information from all women age 15-49. These women were asked questions on the following main topics:

- ♣ Background characteristics (age, residential history, education, literacy, religion, and ethnicity)
- ♣ Reproductive history for the last 6 years
- ♣ Preventive malaria treatment for the most recent birth
- ♣ Prevalence and treatment of fever among children under age 5
- ♣ Knowledge about malaria (symptoms, causes, and how to prevent)
- ♣ Sources of media messages about malaria

The Biomarker Questionnaire was used to record the results of the anaemia and malaria testing of children 6-59 months, as well as the signatures of the fieldworker and the parent or guardian who gave consent. Consent statements were developed for each tool (Household, Woman's, and Biomarker Questionnaires). Further consent statements were formulated for malaria testing, anaemia testing, and treatment of children with positive malaria rapid diagnostic tests (RDTs). Verbal informed consent was sought from eligible respondents before the administration of the Household Questionnaire and the Individual Questionnaire or the collection of biomarker data. For children age 6-59 months eligible for anaemia and malaria testing, consent was obtained from a parent or guardian, or other responsible adult prior to testing. The survey staff recorded their unique identification number and signed to indicate that the consent procedure was properly administered, and whether or not the respondent provided their consent.

1.4 ANAEMIA AND MALARIA TESTING

Blood samples for biomarker testing were collected by finger- or heel-prick from children age 6-59 months. Each field team included two laboratory technicians who carried out the anaemia and malaria testing and prepared the blood smears. A nurse on each field team provided malaria medications for children who tested positive for malaria, in accordance with the approved treatment protocols. The field laboratory technicians requested informed consent for each test from the child's parent or guardian before the blood samples were collected, according to the protocols approved by the National Health Sciences Research Committee in Malawi.

Anaemia testing. A single-use, retractable, spring-loaded, sterile lancet was used to make a finger- or heel-prick. A drop of blood from this site was then collected in a microcuvette. Haemoglobin analysis was carried out on site using a battery-operated, portable HemoCue® analyser, which produces a result in less than one minute. Results were given to the child's parent or guardian verbally and in writing. Parents of children with a haemoglobin level under 8 g/dl were advised to take the child to a health facility for follow-up care and were given a referral letter with the haemoglobin reading to show to staff at the health facility. Results of the anaemia test were recorded on the Biomarker Questionnaire and on a brochure left in the household that also contained information on the causes and prevention of anaemia.

Malaria testing using a rapid diagnostic test (mRDT). Using the same finger- or heel prick that was used for anaemia testing, another drop of blood was tested immediately using the SD BIOLINE Malaria Ag P.f/Pan rapid diagnostic test (RDT). This qualitative test detects the histidine-rich protein II (HRP-II) antigen of *Plasmodium falciparum* and common *Plasmodium* lactate dehydrogenase (pLDH) of *Plasmodium* species in human whole blood (Standard Diagnostics, Inc.). The parasite, transmitted by a mosquito, is the major cause of malaria in Malawi. The diagnostic test includes a disposable sample applicator that comes in a standard package. A tiny volume of blood is captured on the applicator and placed in the well of the testing device. All field laboratory technicians were trained to perform the RDT in the field, in accord with manufacturer's instructions. RDT results were available in 20 minutes and recorded as either positive or negative, with faint test lines considered positive. As with anaemia testing, malaria RDT results were provided to the child's parent or guardian in oral and written form and were recorded on the Biomarker Questionnaire.

Children who tested positive for malaria were offered a full course of medicine according to standard procedures for uncomplicated malaria treatment in Malawi. To ascertain the correct dose, nurses on each field team were trained to use treatment guidance charts and to ask about any medications the child might already be taking. The nurses were also trained to identify signs and symptoms of severe malaria. The nurses provided the age-appropriate dose of artemisinin-based combination therapy (ACT) along with instructions on how to administer the medicine to the child. They also directly observed administration of the first dose.

Malaria testing using blood smears. In addition to the RDT, thick and thin blood smears were prepared in the field. Each blood smear slide was given a bar code label, with a duplicate affixed to the Biomarker Questionnaire. An additional copy of the bar code label was affixed to a blood sample transmittal form to track the blood samples from the field to the laboratory. The slides were dried in a dust-free environment and stored in slide boxes. The thick and thin blood smear

slides were collected regularly from the field, along with the completed Biomarker Questionnaires, and transported to the laboratory for logging and microscopic reading. The smears were stained with Giemsa stain and examined to determine the presence of Plasmodium infection and parasite density. All stained slides were read by two independent microscopists. Slides with discordant microscopy results were reanalysed by a third microscopist for final validation.

The microscopic results were quality checked by internal and external quality control processes. Internal quality control consisted of an independent microscopist who read 5% of all slides in the study. External quality control was conducted through the Malawi College of Medicine Laboratory where 10% of samples were independently read. Ten percent of DBS were also transported together with blood smears to molecular lab at college of medicine for external quality assurance.

1.5 TRAINING OF FIELD STAFF

A 3-week training for field staff took place from 12th to 30th April, 2021, at the Cross Roads Hotel in Lilongwe. Overall, 60 people attended the training, including 10 field supervisors, 30 interviewers, and 20 laboratory technicians. NMCP and NSO staff members led the training and served as the supervisory team for the fieldwork practice.

During the first week of training, field supervisors and interviewers focused on how to fill out the Household and Woman's questionnaires, conduct mock interviews, and use appropriate interviewing techniques.

During the second week, two parallel training sessions were organized: one for the interviewers and team supervisors and the other for laboratory technicians. The training of interviewers and field supervisors focused on the use of the CAPI application for filling out the Household, Woman's and Biomarker questionnaires, assigning households to interviewers, and transferring data from completed questionnaires to the central data processing centre. Throughout the training, quizzes were administered to assess how well the participants absorbed the training materials, both on using the paper questionnaire and on using the CAPI application as a data collection tool.

Training of laboratory technicians focused on preparing blood samples to test for anaemia and malaria. The training involved presentations, discussion, and actual testing for anaemia and malaria. The technicians were trained to identify children eligible for testing, administer informed consent, conduct the anaemia and malaria rapid tests, make thick and thin blood smears, and administer the correct treatment protocols. They were also trained to store the blood slides, record test results on the Biomarker Questionnaire, and provide the results to the parents or guardians of the children tested. Finally, technicians were trained on how to record children's anaemia and malaria results on the respective brochures and how to fill in the referral slip for any child who was found to be severely anaemic. To help put the importance of the 2021 MMIS into context for the trainees, the training also included presentations given by NMCP staff on epidemiology of malaria and Malawi-specific policies and programs on malaria. All participants took part in 3-day field practice exercises in Lilongwe.

1.6 FIELDWORK

Ten teams were organised for field data collection. Each team consisted of one field supervisor, three health professionals to interview and administer treatment, two laboratory technicians to conduct biomarker testing, and two drivers. The field staff also included national coordinators who collected slides from the field teams and delivered them to the National Reference Health Laboratory. Field data collection for the 2021 MMIS started on 3 May 2021. For maximum supervision, all ten teams were visited by national monitors, at least once a week. Fieldwork was completed on 30 June 2021. Note also that this survey was done in a Covid 19 era and all necessary preventive measures as laid down by National Covid Task Force were strictly followed

1.7 DATA PROCESSING

Data for the 2021 MMIS were collected through questionnaires programmed onto the CAPI application. The CAPI were programmed by NSO and loaded with the Household, Biomarker, and Woman's Questionnaires in all three major languages. Using the cloud, the field supervisors transferred data on a daily basis to a central location for data processing in Lilongwe. To facilitate communication and monitoring, each field worker was assigned a unique identification number. NSO provided technical assistance for processing the data using the Censuses and Surveys Processing (CSPRO) system for data editing, cleaning, weighting, and tabulation. In the central office, data received from the field teams' CAPI applications were registered and checked for any inconsistencies. Data editing and cleaning included an extensive range of structural and internal consistency checks. Any anomalies were communicated to team (field) supervisors so that the data processing teams could resolve data discrepancies. The corrected results were maintained in master CSPRO data files at NSO and used for analysis in producing tables for the final report.

1.8 RESPONSE RATES

Table 1.1 shows that of the 3,750 households selected for the sample, 3,726 were occupied at the time of fieldwork. Among the occupied households, 3,711 were successfully interviewed, yielding a total household response rate of 99.6%. In the interviewed households, 3,765 eligible women were identified as eligible for individual interview, and 3,707 women were successfully interviewed, yielding a response rate of 99%.

Table 1:1 Results of the households and individual interviews

Number of households, number of interviews and response rates according to residence (unweighted), Malawi MMIS 2021

<i>Results</i>	Residence		
	Urban	Rural	Total
Household Interviews			
Household Selected	1500	2250	3750
Household Occupied	1489	2237	3726
Household Interviewed	1481	2230	3711
<i>Household Response Rate*</i>	99.5	99.7	99.6
Interviews with Women aged 15 - 49			
Number of eligible women	1654	2111	3765
Number of eligible women interviewed	1635	2074	3709
<i>Eligible women response rate**</i>	98.9	98.2	98.5
*Households interviewed/Households occupied			
**Respondents interviewed/Eligible respondents			

CHARACTERISTICS OF HOUSEHOLDS AND WOMEN 2

Key Findings

- **Drinking Water:** Majority of urban households (97%) have access to an improved source of drinking water while 90% of rural households have access.
- **Sanitation:** Only 9% of the households use an improved toilet facility; 16% of the households have a toilet facility that would be classified as improved toilet facility, 75% use an unimproved toilet facility and 6% practice open defecation.
- **Household Wealth:** 24% of households in the Northern region are in the middle wealth quintile while majority of households in the Central and Southern regions (30% and 33% respectively) are in the lowest wealth quintile.
- **Electricity:** 93% of rural households do not have electricity
- **Bank Account/Village Savings:** 42% of households have a member with a bank account (80% in urban areas and 35% in rural areas).
- **Literacy:** Overall, younger women are more likely to be literate than older women. Eighty-eight percent of women age 15-19 are literate compared with 40% of women age 45-49.

Information on the socioeconomic characteristics of the household population in the 2021 MMIS provides a context to interpret demographic and health indicators and can furnish an approximate indication of the representativeness of the survey. In addition, this information sheds light on the living conditions of the population of Malawi.

This chapter presents information on source of drinking water, sanitation, wealth, ownership of durable goods and composition of the household population. In addition, the chapter presents characteristics of the survey respondents such as age, education and literacy. Socioeconomic characteristics are useful for understanding the factors that affect use of health services and other health behaviours related to malaria control.

2.1 DRINKING WATER SOURCES AND TREATMENT

Improved sources of drinking water

Includes piped water, public taps, standpipes, tube wells, boreholes, protected dug wells, springs, and rainwater. Households using bottled water for drinking are classified as using an improved source only if their water for cooking and handwashing is from an improved source.

Sample: Households

Improved sources of water protect against outside contamination so water is more likely to be safe to drink. In Malawi, 91% of households have access to an improved source of drinking water (**Table 2.1**). Ninety-seven percent of urban households and 90% of rural households have access to improved water sources. Urban and rural households rely on different sources of drinking water. Forty-six percent of urban households have piped water in their dwelling or yard, which accounts for the largest percentage of improved water sources for urban households (**Figure 2.1**). In contrast, rural households with access to improved sources of drinking water rely mainly on tube wells or boreholes (70%). Ninety-three percent of urban and 74% of rural households have water on the premises or travel less than 30 minutes to fetch drinking water (**Table 2.1**).

Trends: The proportion of households obtaining water from improved sources increased slightly from 81% in the 2012 MMIS to 83% in the 2014 MMIS, 86% in the 2017 MMIS and then 91% in 2021 MMIS. The increase occurred in both urban and rural households, from 93% in 2012 to 97% in 2021 in urban areas and from 79% in 2012 to 90% in 2021 in rural areas (**Figure 2.2**).

Figure 2.1: Household drinking water by residence

Percent distribution of households by source of drinking water

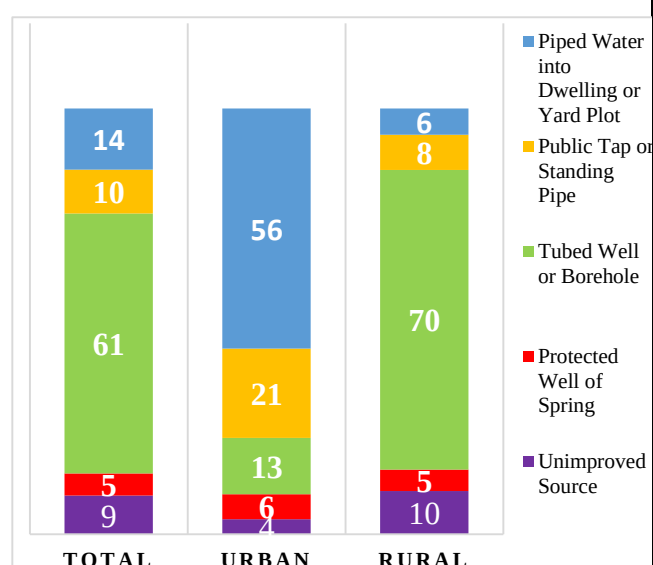
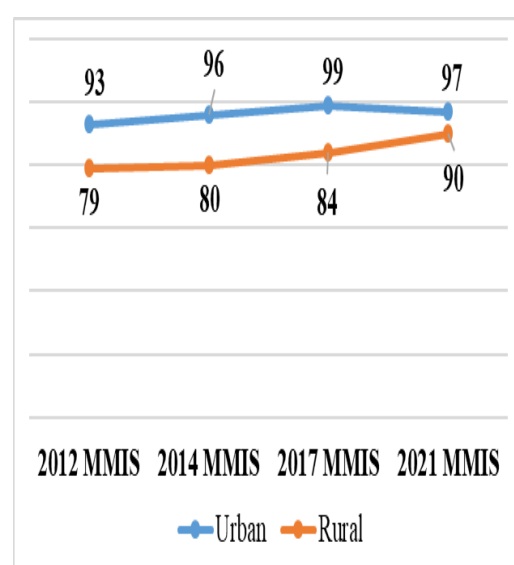


Figure 1.2: Trends in sources of drinking water by residence

Percent obtaining drinking water from improved sources



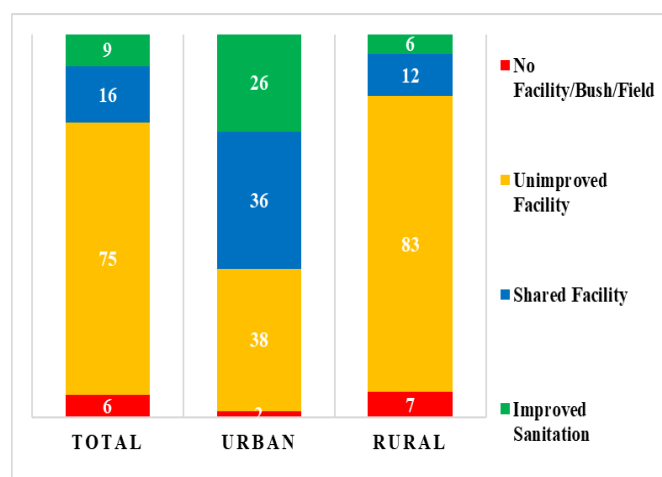
2.2 SANITATION

Improved toilet facilities

Include any non-shared toilet of the following types: flush/pour flush toilets to piped sewer systems, septic tanks, and pit latrines; ventilated improved pit (VIP) latrines; pit latrines with slabs; and composting toilets

Sample: Households

Nationally, only 9% of households use an improved sanitation facility. Ninety-one percent of households use unimproved sanitation, including 16% of the households that have a toilet facility that would be classified as improved if it were not shared with other households, 75% that use an unimproved toilet facility, and 6% that practice open defecation (**Figure 2.3**). Households in urban areas (26%) are more likely than rural households (6%) to use improved facilities. The most commonly used improved toilet facility is the pit latrine with a slab (8%) (**Table 2.2**).



Note: Totals may not add to 100 due to rounding.

Figure 2.3 Household toilet facilities by residence

Percent distribution of households by type of toilet facilities

2.3 HOUSING CHARACTERISTICS

The 2021 MMIS collected data on household features such as access to electricity, flooring material, number of sleeping rooms, and types of fuel used for cooking. The responses to these questions, along with information on ownership of household durable goods, contribute to the creation of the household wealth index and provide information that may be relevant for other health indicators.

Exposure to cooking smoke, especially that produced from solid fuels such as charcoal and firewood, is potentially harmful to health. Both urban and rural households rely on solid cooking fuels, but there are differences in the type of solid fuels they use. The percentage of households using charcoal for cooking is about 5 times higher in urban households than in rural households (68% versus 14%, respectively), while rural households are more likely to use firewood than urban households (85% versus 22%, respectively). (**Table 2.3**).

Overall, at least 1 in 7 (14%) households in Malawi has access to electricity. Fifty-two percent of urban households but only 7% of rural households have access to electricity.

The most common flooring material in Malawi is earth or sand (67%). Cement is the second most common material (30%). By residence, the most common choice in urban areas is cement (72%), while in rural areas it is earth or sand (75%).

The number of rooms a household uses for sleeping is an indicator of socioeconomic level and crowding in the household; crowding can facilitate the spread of diseases. In Malawi, most households (38%) of households use two rooms for sleeping, while 31% use either a single room or more than three rooms for sleeping. In urban areas, 37% use two rooms, while in rural areas 38% use two rooms for sleeping (**Table 2.3**).

Household Durable Goods

Data displayed in **Table 2.4** presents information on ownership of household effects, means of transport, agricultural land and animals, and bank accounts. Urban households are more likely than rural households to own a radio (53% versus 26%), television (43% versus 6%), mobile phone (87% versus 56%), and car or truck (12% versus 1%). Urban households are also more likely than rural households to own a bank account (80% versus 35%). In contrast, rural households are more likely than urban households to own agricultural land (82% versus 32%), farm animals (51% versus 25%), a bicycle (31% vs 30%), and a torch (82% versus 66%).

2.4 HOUSEHOLD WEALTH

Wealth index

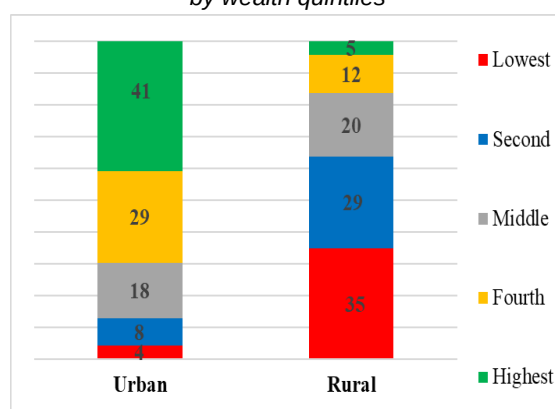
Households are given scores based on the number and kinds of consumer goods they own, ranging from a television to a bicycle or car, and housing characteristics such as source of drinking water, toilet facilities, and flooring materials. These scores are derived using principal component analysis. National wealth quintiles are compiled by assigning the household score to each usual (de jure) household member, ranking each person in the household population by their score, and then dividing the distribution into five equal categories, each with 20% of the population.

Sample: Households

There are substantial variations in the distribution of wealth quintiles at residential and regional levels. Rural residences have the highest percentage of population in the lowest quintile (35%) compared with urban ones (4%), while 41% of the urban population is in the highest quintile compared with 5% of the rural population (Figure 2.4). At the regional level, the Southern region has the highest percentage of population in the lowest quintile (33%) as well as the lowest percentage of the population in the highest wealth quintile (8%). The Northern region has the highest percentage of population in the highest wealth quintile (16%) compared with Central (11%) and Southern (8%) regions (Table 2.5).

Figure 2.4 Household wealth by residence

Percent distribution of de jure population by wealth quintiles



2.5 HOUSEHOLD POPULATION AND COMPOSITION

Household

A person or group of related or unrelated persons who live together in the same dwelling unit(s), who acknowledge one adult male or female as the head of the household, who share the same housekeeping arrangements, and who are considered a single unit.

De facto population

All persons who stayed in the selected households the night before the interview (whether usual residents or visitors)

De jure population

All persons who are usual residents of the selected households, whether or not they stayed in the household the night before the interview

How data are calculated

All tables are based on the de facto population, unless specified otherwise.

In the 2021 MMIS, 15,956 people stayed overnight in 3,711 households. The population sex ratio is 95 males per 100 females. By residence, there were 97 males per 100 females in urban areas and 94 males per 100 females in rural areas. Eighty-four percent of the population lives in rural areas (**Table 2.6**).

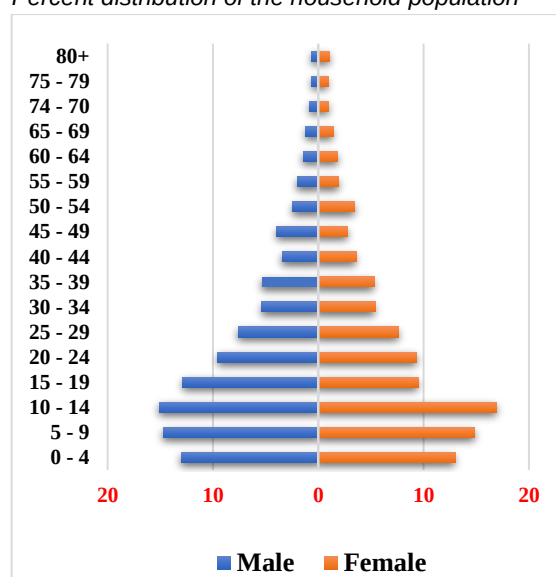
Age and sex are important demographic variables and are the primary basis of demographic classification. **Table 2.6** shows the distribution of the de facto household population in the 2017 MMIS by 5-year age groups, according to sex and residence.

The population in **Figure 2.5** shows the population distribution by sex and by five-year age groups. The broad base of the pyramid shows that the Malawi population is young, which is typical of developing countries with a higher fertility rate and low life expectancy. Forty-four percent of population is under the age 15, almost half of the population (52%) is between the age of 15 and 64, and only 4% of the population is aged 65 and above (**Table 2.6**).

On average, household in Malawi consist of 4.3 persons (**Table 2.7**). Men are predominantly the head of households in Malawi (72%). The proportion of households headed by women is slightly higher in the rural areas than in urban areas (29% versus 23%).

Figure 2.5 Population pyramid

Percent distribution of the household population



2.6 BACKGROUND CHARACTERISTICS OF WOMEN RESPONDENTS

Table 2.8 shows by background characteristics the weighted and unweighted numbers and the weighted percentage distributions of women age 15-49 who were interviewed in the 2021 MMIS, by background characteristics. More than half of the respondents (60%) are between age 15 and 30, which reflects the youthful population. The majority of respondents are Christians (73%), with Catholic and CCAP forming the largest composition (28%), and 17% are Muslims.

The Chewas form the largest ethnic group (34%). Eighty-one percent of respondents live in rural areas. The largest percentage of respondents lives in the Southern region (46%) followed by the Central region (42%) and the Northern region (12%).

2.7 EDUCATIONAL ATTAINMENT OF WOMEN

Studies have consistently shown that educational attainment has a strong effect on health behaviours and attitudes. Generally, the higher the level of education a woman has attained, the more knowledgeable she is about both the use of health facilities and health management for herself and for her children.

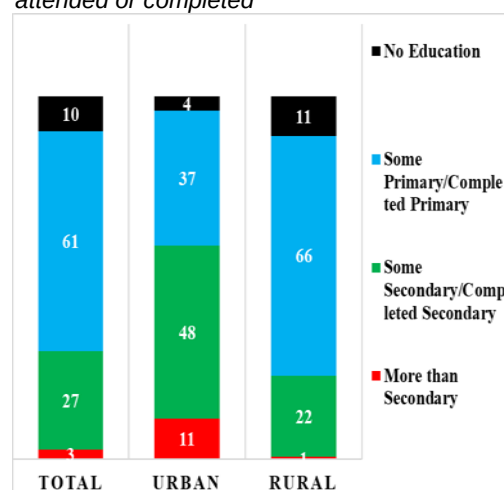
In general, 10% of women in Malawi have no education (**Figure 2.6**). Ten percent of women have completed secondary education or attended higher than secondary school. Overall, women have completed a median of 7 years of education (**Table 2.9**).

Patterns by background characteristics

- Eleven percent of women in rural areas have no education compared with 4% in urban areas.
- At the regional level, the Southern region recorded the highest percentage of women with no education (11%), while the Northern region had the lowest percentage (3%) of women with no education.
- The percentage of respondents with no education decreases with increasing wealth quintile, from 16% in the lowest quintile to 2% in the highest quintile.

Figure 2.6 Education of survey respondents

Percent distribution of women age 15-49 by highest level of schooling attended or completed



2.8 LITERACY OF WOMEN

Literacy

Respondents who have attended higher than secondary school are assumed to be literate. All other respondents were given a sentence to read, and they were considered to be literate if they could read all or part of the sentence.

Sample: Women age 15-49

The results show that only 3% of women have higher than secondary education. Amongst those with secondary education or lower including those with no education, 32% can read the whole sentence, 42% can read part of the sentence and 23% cannot read at all. The results show that, overall, 76% of women age 15-49 are literate. (Table 2.10).

Patterns by background characteristics

- Literacy is much higher amongst the youngest women age 15-19 (88%) and decreases steadily with age to 40% amongst the oldest women (45-49) (Table 2.10).
- Literacy varies by place of residence as 91% of women in urban areas are literate, compared with 73% of rural women.
- Regional differences in literacy are notable; literacy is highest among women in the Northern region (87%) and lowest among women in the Central region (73%).
- By wealth, literacy ranges from 60% among women in the lowest wealth quintile to 94% among women in the highest quintile.

LIST OF TABLES

For detailed information on household population and housing characteristics, see the following tables:

- **Table 2.1 Household drinking water**
- **Table 2.2 Household sanitation facilities**
- **Table 2.3 Household characteristics**
- **Table 2.4 Household possessions**
- **Table 2.5 Wealth quintiles**
- **Table 2.6 Household population by age, sex, and residence**
- **Table 2.7 Household composition**
- **Table 2.8 Background characteristics of respondents**
- **Table 2.9 Women's educational attainment**
- **Table 2.10 Women's literacy**

Table 2.1 Household drinking water

Percent distribution of households and de jure population by source of drinking water and by time to obtain drinking water, according to residence, Malawi MIS 2021

Characteristic	Households			Population		
	Urban	Rural	Total	Urban	Rural	Total
Source of drinking water						
Improved source	96.5	90.0	91.1	96.3	89.8	90.8
Piped into dwelling/yard/plot	45.7	3.5	10.4	47.5	3.9	10.8
Piped to neighbor	10.7	2.7	4.0	10.0	2.7	3.9
Public tap/standpipe	21.0	8.2	10.3	20.9	7.9	9.9
Tubewell/borehole	13.2	70.4	61.1	12.2	70.8	61.5
Protected dug well	4.8	4.6	4.6	4.6	4.0	4.1
Protected spring	0.8	0.4	0.5	0.9	0.4	0.5
Rainwater	0.0	0.1	0.1	0.0	0.1	0.1
Bottled water, improved source for cooking/handwashing ¹	0.3	0.1	0.1	0.2	0.1	0.1
Unimproved source	3.5	10.0	8.9	3.7	10.2	9.2
Unprotected dug well	2.8	5.7	5.2	3.0	6.0	5.5
Unprotected spring	0.5	1.5	1.4	0.4	1.5	1.3
Tanker truck/cart with small tank	0.0	0.0	0.0	0.0	0.0	0.0
Surface water	0.3	2.7	2.3	0.3	2.7	2.3
Bottled water, unimproved source for cooking/handwashing ¹	0.0	0.0	0.0	0.0	0.0	0.0
Other source	0.0	0.0	0.0	0.0	0.0	0.0
Total	100.0	100.0	100.0	100.0	100.0	100.0
Time to obtain drinking water (round trip)						
Water on premises ²	63.2	15.8	23.5	63.9	16.0	23.6
Less than 30 minutes	29.6	58.2	53.6	28.9	57.5	52.9
30 minutes or longer	7.0	25.1	22.2	7.1	25.7	22.7
Don't know	0.2	0.8	0.7	0.1	0.8	0.7
Total	100.0	100.0	100	100.0	100.0	100.0
Number of households/population	607	3,104	3711	2,546.5	13,410	15,956

¹ Households using bottled water for drinking are classified as using an improved or unimproved source according to their water source for cooking and handwashing.

² Includes water piped to a neighbor

Table 2.2 Household sanitation facilities

Percent distribution of households and de jure population by type of toilet/latrine facilities, according to residence, Malawi MIS 2021

Type and location of toilet/latrine facility	Households			Population		
	Urban	Rural	Total	Urban	Rural	Total
Improved sanitation	25.9	5.5	8.8	23.5	4.7	7.7
Flush/pour flush to piped sewer system	0.3	0.0	0.0	0.1	0.0	0.0
Flush/pour flush to septic tank	0.3	0.0	0.0	0.3	0.0	0.0
Flush/pour flush to a pit latrine	0.1	0.1	0.1	0.2	0.1	0.1
Ventilated improved pit (VIP) latrine	0.3	0.6	0.5	0.2	0.6	0.5
Pit latrine with a slab	24.8	4.6	7.9	22.6	4.0	7.0
Composting toilet	0.1	0.2	0.1	0.1	0.1	0.1
Unimproved sanitation	74.1	94.5	91.2	76.2	95.0	92.0
Shared facility¹	36.4	11.8	15.8	39.5	12.5	16.8
Flush/pour flush to piped sewer system	3.3	0.0	0.5	3.1	0.0	0.5
Flush/pour flush to septic tank	12.9	0.1	2.2	13.6	0.2	2.3
Flush/pour flush to a pit latrine	0.5	0.2	0.3	0.6	0.2	0.3
Ventilated improved pit (VIP) latrine	1.1	1.5	1.4	1.2	1.5	1.4
Pit latrine with a slab	18.6	9.9	11.3	21.0	10.6	12.2
Composting toilet	0.0	0.1	0.0	0.0	0.1	0.0
Unimproved facility	37.8	82.7	75.3	36.7	82.6	75.3
Flush/pour flush not to sewer/ septic tank/pit latrine	0.2	0.0	0.0	0.3	0.0	0.0
Pit latrine without slab/open pit	36.0	75.4	69.0	35.2	76.2	69.6
Bucket	0.0	0.1	0.0	0.0	0.1	0.1
Hanging toilet/hanging latrine	0.0	0.2	0.2	0.1	0.4	0.3
Open defecation (no facility/bush/field)	1.6	6.9	6.1	1.2	5.9	5.1
Total	100.0	100.0	100.0	100.0	100.0	100.0
Number of households/population						
Number of households/population	607	3,104	3,711	2,554	13,441	15,995
Number of households/population with a toilet/latrine facility	598	2,888	3,486	2,516	12,621	15,138

¹ Facilities that would be considered improved if they were not shared by two or more households

Table 2.2 Household sanitation facilities

Percent distribution of households and de jure population by type of toilet/latrine facilities, according to residence, Malawi MIS 2021

Type and location of toilet/latrine facility	Households			Population		
	Urban	Rural	Total	Urban	Rural	Total
Improved sanitation	25.9	5.5	8.8	23.5	4.7	7.7
Flush/pour flush to piped sewer system	0.3	0.0	0.0	0.1	0.0	0.0
Flush/pour flush to septic tank	0.3	0.0	0.0	0.3	0.0	0.0
Flush/pour flush to a pit latrine	0.1	0.1	0.1	0.2	0.1	0.1
Ventilated improved pit (VIP) latrine	0.3	0.6	0.5	0.2	0.6	0.5
Pit latrine with a slab	24.8	4.6	7.9	22.6	4.0	7.0
Composting toilet	0.1	0.2	0.1	0.1	0.1	0.1

Unimproved sanitation	74.1	94.5	91.2	76.2	95.0	92.0
Shared facility¹	36.4	11.8	15.8	39.5	12.5	16.8
Flush/pour flush to piped sewer system	3.3	0.0	0.5	3.1	0.0	0.5
Flush/pour flush to septic tank	12.9	0.1	2.2	13.6	0.2	2.3
Flush/pour flush to a pit latrine	0.5	0.2	0.3	0.6	0.2	0.3
Ventilated improved pit (VIP) latrine	1.1	1.5	1.4	1.2	1.5	1.4
Pit latrine with a slab	18.6	9.9	11.3	21.0	10.6	12.2
Composting toilet	0.0	0.1	0.0	0.0	0.1	0.0
Unimproved facility	37.8	82.7	75.3	36.7	82.6	75.3
Flush/pour flush not to sewer/ septic tank/pit latrine	0.2	0.0	0.0	0.3	0.0	0.0
Pit latrine without slab/open pit	36.0	75.4	69.0	35.2	76.2	69.6
Bucket	0.0	0.1	0.0	0.0	0.1	0.1
Hanging toilet/hanging latrine	0.0	0.2	0.2	0.1	0.4	0.3
Open defecation (no facility/bush/field)	1.6	6.9	6.1	1.2	5.9	5.1
Total	100.0	100.0	100.0	100.0	100.0	100.0
Number of households/population						
Number of households/population	607	3,104	3,711	2,554	13,441	15,995
Number of households/population with a toilet/latrine facility	598	2,888	3,486	2,516	12,621	15,138

Table 2.3 Household characteristics

[illegible]

Rooms used for sleeping

One	27.6	32.0	31.2	17.1	23.1	22.1
Two	36.5	37.7	37.5	37.4	38.1	37.9
Three or more	35.9	30.4	31.3	45.5	38.9	39.9
Total	100.0	100.0	100.0	100.0	100.0	100.0

Cooking fuel

Electricity	9.8	0.3	1.8	8.8	0.2	1.6
LPG/natural gas/biogas	0.4	0.0	0.0	0.3	0.0	0.0
Kerosene	0.0	0.0	0.0	0.0	0.0	0.0
Coal/lignite	0.0	0.0	0.0	0.0	0.0	0.0
Charcoal	67.5	13.7	22.5	67.9	12.4	21.2
Wood	22.1	85.4	75.0	22.6	86.8	76.5
Straw/shrubs/grass	0.0	0.2	0.2	0.0	0.2	0.2
Agricultural crop	0.3	0.3	0.3	0.4	0.3	0.3
Animal dung	0.0	0.0	0.0	0.0	0.0	0.0
Other fuel	0.0	0.0	0.0	0.0	0.0	0.0
No food cooked in household	0.0	0.1	0.1	0.0	0.1	0.1
Total	100.0	100.0	100.0	100.0	100.0	100.0

Percentage using solid fuel for cooking ¹	89.9	99.7	98.1	90.9	99.7	98.3
Percentage using clean fuel for cooking ²	10.1	0.3	1.9	9.1	0.3	1.7
Total	100.0	100.0	100.0	100.0	100.0	100.0
Number of households/population	607	3,104	3,711	2,547	13,410	15,956

LPG = Liquefied petroleum gas

[1] Includes coal/lignite, charcoal, wood, straw/shrubs/grass, agricultural crops, and animal dung [LIST ANY ADDITIONAL CATEGORIES INCLUDED IN THE SURVEY QUESTIONNAIRE]

[2] Includes electricity and LPG/natural gas and biogas [LIST ANY ADDITIONAL CATEGORIES INCLUDED IN THE SURVEY QUESTIONNAIRE]

Table 2.4 Household possessions

Percentage of households possessing various household effects, means of transportation, agricultural land and livestock/farm animals, according to residence, Malawi MIS 2021

Possession	Residence		Total
	Urban	Rural	
Household effects			
Radio	52.7	26.2	30.6
Television	43.4	6.3	12.3
Mobile_phone	87.2	56.3	61.3
Telephone	2.9	1.0	1.3
Computer	14.8	1.5	3.7
Refrigerator	27.4	2.1	6.2
Koloboyi	2.2	2.2	2.2
Paraffin_lamp	2.2	2.3	2.3
Bed_matress	67.3	24.3	31.4

	Torch	65.5	81.6	79.0
	Sofa	42.8	8.1	13.7
Means of transport				
	Bicycle	29.8	31.2	31.0
	Animal drawn cart	0.7	1.9	1.7
	Motorcycle/scooter	5.3	4.8	4.9
	Car/truck	11.9	1.3	3.0
	Boat with a motor	0.2	0.3	0.3
Ownership of agricultural land				
		31.5	82.2	73.9
Ownership of farm animals¹				
		25.4	51.2	47.0
Number of households		607	3,104	3,711

¹ Cows, bulls, other cattle, horses, donkeys, mules, goats, sheep, chickens, or other poultry

Table 2.5 Wealth quintiles

Percent distribution of the de jure population by wealth quintiles and the Gini Coefficient, according to residence and region, Malawi MIS 2021

Residence/region	Wealth quintile					Total	Number of persons	Gini Coefficient
	Lowest	Second	Middle	Fourth	Highest			
Residence								
Urban	4.3	8.4	17.6	29.1	40.7	100.0	2,547	
Rural	34.7	29.1	20.1	11.7	4.5	100.0	13,410	
Region								
Northern Region	17.1	21.7	23.6	21.3	16.3	100.0	1,938	
Central Region	30.3	26.9	17.5	14.2	11.0	100.0	6,856	
Southern Region	32.8	25.8	20.7	12.8	7.8	100.0	7,162	
Total	29.8	25.8	19.7	14.4	10.2	100.0	15,956	

Table 2.6 Household population by age, sex, and residence

Percent distributions of the de facto household population by various age groups and percentage of the de facto household population age 10-19, according to sex and residence, Malawi MIS 2021

Age	Urban			Rural			Total		
	Male	Female	Total	Male	Female	Total	Male	Female	Total
<5	10.7	11.6	11.1	13.5	13.3	13.4	13.0	13.0	13.0
5-9	13.2	13.0	13.1	15.0	15.2	15.1	14.7	14.9	14.8
10-14	12.6	14.6	13.6	15.6	17.4	16.5	15.1	16.9	16.0
15-19	10.9	9.5	10.2	13.2	9.5	11.3	12.8	9.5	11.1
20-24	12.0	11.2	11.6	9.1	9.0	9.0	9.6	9.3	9.4
25-29	8.8	10.5	9.6	7.4	7.1	7.2	7.6	7.6	7.6
30-34	7.5	7.8	7.6	5.0	4.9	5.0	5.4	5.4	5.4
35-39	6.1	6.3	6.2	5.2	5.1	5.1	5.3	5.3	5.3
40-44	5.2	4.5	4.9	3.1	3.4	3.3	3.4	3.6	3.5
45-49	4.5	2.6	3.5	3.8	2.8	3.3	3.9	2.8	3.3
50-54	2.5	3.4	3.0	2.4	3.5	3.0	2.4	3.5	3.0
55-59	1.8	1.4	1.6	2.0	2.0	2.0	2.0	1.9	2.0
60-64	1.8	1.7	1.8	1.4	1.9	1.6	1.4	1.8	1.6
65-69	1.0	0.8	0.9	1.2	1.5	1.4	1.2	1.4	1.3
70-74	0.8	0.5	0.7	0.8	1.1	0.9	0.8	1.0	0.9
75-79	0.4	0.2	0.3	0.7	1.1	0.9	0.6	0.9	0.8
80 +	0.1	0.4	0.3	0.7	1.2	1.0	0.6	1.1	0.9
Total	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
Dependency age groups									
0-14	36.4	39.2	37.8	44.1	45.9	45.0	42.8	44.8	43.8
15-64	61.1	58.8	59.9	52.5	49.2	50.8	53.9	50.8	52.3
65+	2.5	2.0	2.2	3.4	4.9	4.2	3.3	4.4	3.9
Total	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
Child and adult populations									
0-17	43.1	44.8	44.0	52.9	51.5	52.2	51.3	50.5	50.9
18+	56.9	55.2	56.0	47.1	48.5	47.8	48.7	49.5	49.1
Total	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
Adolescents 10-19	23.5	24.1	23.8	28.8	26.9	27.8	28.0	26.4	27.2
Number of persons	1,251	1,296	2,547	6,510	6,900	13,410	7,761	8,195	15,956

Table 2.7 Household composition

Percent distribution of households by sex of head of household and by household size and mean size of households, according to residence, Malawi MIS 2021

Characteristic	Residence		Total
	Urban	Rural	
Household headship			
Male	77.3	70.9	72.0

Female	22.7	29.1	28.0
Total	100.0	100.0	100.0
Number of usual members			
1	9.7	7.0	7.5
2	11.9	11.4	11.5
3	16.9	20.4	19.8
4	18.9	19.2	19.2
5	17.9	15.3	15.7
6	11.9	12.1	12.0
7	7.7	7.5	7.6
8	3.2	3.7	3.6
9+	2.0	3.4	3.2
Total	100.0	100.0	100.0
Mean size of households	4.2	4.3	4.3
Number of households	607	3,104	3,711

Note: Table is based on de jure household members, i.e., usual residents.

Table 2.8 Background characteristics of respondents

Percent distribution of women age 15-49 by selected background characteristics, Malawi MIS 2021

Background characteristic	Women		
	Weighted percent	Weighted number	Unweighted number
Age			
15-19	22.4	830	782
20-24	21.1	781	811
25-29	17.0	629	642
30-34	12.4	458	475
35-39	12.3	455	444
40-44	8.7	324	330
45-49	6.2	231	225
Religion			
Catholic	15.9	591	603
CCAP	12.3	455	645
Anglican	2.6	98	99
Seventh Day/Baptist	6.8	250	247
Other Christian	35.4	1314	1308
Muslim	17.3	643	452
No religion	0.8	29	19
Other	8.9	329	336
Ethnic group			
Chewa	33.7	1248	1002
Tumbuka	8.6	318	841
Lomwe	20.3	753	576
Tonga	1.4	53	120
Yao	15.2	566	411
Sena	1.6	58	75
Nkhonde	0.8	30	78

Ngoni	11.9	441	353
Other	6.5	243	253
Residence			
Urban	18.9	712	1,654
Rural	81.1	3,052	2,111
Region			
Northern Region	12.0	451	1,289
Central Region	42.1	1,584	1,232
Southern Region	45.9	1,729	1,244
Education			
No education	9.6	355	240
Primary	60.5	2,245	1,962
Secondary	27.3	1,011	1,307
More than secondary	2.7	99	200
Wealth quintile			
Lowest	28.3	1,066	696
Second	25.1	945	683
Middle	20.4	766	763
Fourth	14.2	536	746
Highest	12.0	452	877
Total 15-49	100.0	3,764	3,765

Note: Education categories refer to the highest level of education attended, whether or not that level was completed.

na = Not applicable

Table 2.9 Educational attainment of survey respondents

Percent distribution of women age 15-49 by highest level of schooling attended or completed, and median years completed, according to background characteristics, Malawi MIS 2021

Background characteristic	Highest level of schooling							Total	Median years completed	Number of women
	No education	Some primary	Completed primary ¹	Some secondary	Completed secondary ²	More than secondary	Missing			
Age	9.6	47.4	13.1	17.3	9.9	2.7	0.0	100.0	7.0	830
15-24	3.3	48.2	15.5	21.9	9.9	1.3	0.0	100.0	8.0	781
15-19	2.0	50.0	16.8	26.4	4.7	0.2	0.0	100.0	8.0	629
20-24	4.6	46.3	14.1	17.2	15.4	2.4	0.0	100.0	7.0	458
25-29	9.5	38.3	14.4	18.0	14.5	5.3	0.0	100.0	7.0	455
30-34	7.2	54.6	9.0	15.0	9.3	4.9	0.0	100.0	6.0	324
35-39	13.6	48.0	13.1	14.7	8.5	2.1	0.0	100.0	5.0	231
40-44	16.8	52.3	9.3	9.7	9.4	2.4	0.0	100.0	10.0	712
45-49	40.1	44.7	6.3	3.6	2.8	2.4	0.0	100.0	7.0	3,052
Residence										
Urban	3.8	26.3	10.7	25.1	22.0	11.1	1.0	100.0	8.0	451
Rural	10.7	51.5	13.4	15.2	7.0	0.6	1.6	100.0	7.0	1,584

Region										
Region 1	2.7	38.2	18.3	21.3	13.8	3.8	1.9	100.0	7.0	1,729
Region 2	10.0	47.9	12.1	16.0	9.3	3.4	1.2	100.0	6.0	1,066
Region 3	10.6	47.8	12.3	16.9	9.2	1.6	1.6	100.0	7.0	945
Wealth quintile										
Lowest	16.4	60.3	11.5	8.7	1.6	0.0	1.6	100.0	7.0	766
Second	12.1	53.7	12.9	14.5	4.9	0.1	1.7	100.0	9.0	536
Middle	5.1	48.0	15.1	19.1	11.2	0.2	1.4	100.0	11.0	452
Fourth	3.2	30.9	16.2	29.2	16.1	2.9	1.5	100.0	0.0	0
Highest	1.9	16.4	8.8	24.4	29.7	18.0	0.8	100.0	0.0	0
Total	9.6	47.4	13.1	17.3	9.9	2.7	0.0	100.0	7.0	830

¹ Completed X grade at the primary level

² Completed Y grade at the secondary level

Table 2.10 Literacy

Percent distribution of women age 15-49 by level of schooling attended and level of literacy, and percentage literate, according to background characteristics, Malawi MIS 2021

Background characteristic	Higher than secondary schooling	No schooling, primary or secondary school					Missing	Total	Percentage literate ¹	Number of women
		Can read a whole sentence	Can read part of a sentence	Cannot read at all	No card with required language	Blind/visually impaired				
Age										
15-24	0.2	40.3	47.4	11.1	1.0	0.0	0.0	100.0	87.9	1,611
15-19	2.4	31.9	47.2	18.5	0.1	0.0	0.0	100.0	81.4	830
20-24	5.3	25.5	46.2	22.1	0.9	0.0	0.0	100.0	77.0	781
25-29	4.9	31.4	39.0	23.9	0.8	0.0	0.0	100.0	75.3	629
30-34	2.1	30.7	37.7	29.5	0.0	0.0	0.0	100.0	70.5	458
35-39	2.4	30.6	33.0	32.2	1.8	0.0	0.0	100.0	66.0	455
40-44	2.4	20.1	17.8	58.8	0.8	0.0	0.0	100.0	40.4	324
45-49	11.2	20.9	58.6	9.0	0.3	0.0	0.0	100.0	90.7	231
Residence										
Urban	0.7	34.2	37.9	26.5	0.8	0.0	0.0	100.0	72.7	712
Rural	3.9	30.3	53.0	12.8	0.1	0.0	0.0	100.0	87.1	3,052
Region										
Northern Region	3.5	29.7	39.8	26.2	0.8	0.0	0.0	100.0	73.0	451
Central Region	1.6	33.8	40.7	23.1	0.8	0.0	0.0	100.0	76.1	1,584
Southern Region	0.0	32.3	28.0	38.7	0.9	0.0	0.0	100.0	60.3	1,729
Wealth quintile										
Lowest	0.1	36.4	34.9	28.2	0.5	0.0	0.0	100.0	71.3	1,066

Second	0.2	35.2	47.5	16.4	0.7	0.0	0.0	100.0	82.9	945
Middle	2.9	30.0	57.9	8.0	1.3	0.0	0.0	100.0	90.8	766
Fourth	18.1	16.0	60.1	5.6	0.1	0.0	0.0	100.0	94.3	536
Highest	2.7	31.6	41.8	23.2	0.7	0.0	0.0	100.0	76.1	452
Total	1.3	36.2	47.3	14.7	0.6	0.0	0.0	100.0	84.7	3,709

¹ Refers to women who attended schooling higher than the secondary level and women who can read a whole sentence or part of a sentence

Key Findings

Ownership of insecticide-treated nets (ITNs):

- Fifty-five percent of households in Malawi own at least one ITN.
- Twenty-one percent of households have at least one ITN for every two people.

Sources of ITNs:

- Forty-four percent of ITNs owned by households were obtained from mass distribution campaigns, 18% from antenatal care visits, 9% from distribution to newborns at the time of birth and 0.1% from shops or markets.

Access to an ITN:

- Sixteen percent of the household population has access to an ITN. This means that at least 1 in 5 people could sleep under an ITN if every ITN in a household were used by up to two people.

Use of ITNs:

- Thirty-seven percent of the household population, 53% of children under 5, and 49% of pregnant women slept under an ITN the night before the survey.
- In households owning at least one ITN, 64% of the household population, 81% of children under 5, and 77% of pregnant women slept under an ITN the previous night.

Intermittent Preventive Therapy (IPTp):

- Fifty-six percent of women age 15-49 with a live birth in the 2 years preceding the survey received at least three doses of SP/Fansidar for prevention of malaria in pregnancy.

This chapter describes the population coverage rates of two key malaria control interventions in Malawi: use of insecticide-treated nets (ITNs) and intermittent preventive treatment in pregnancy (IPTp). Malaria control efforts focus on scaling-up these interventions.

The 2017-2022 Malawi Malaria Control Strategic Plan envisages universal coverage of the population with ITNs through routine distribution and mass campaigns to reduce the burden of malaria. ITNs are routinely given to pregnant women during their first antenatal care visit. They are also given to newborns

delivered at health facilities or at their first visit to a facility or outreach clinic if they did not receive an ITN at birth.

3.1 OWNERSHIP OF INSECTICIDE-TREATED NETS

Ownership of insecticide-treated nets

Households that have at least one insecticide-treated net (ITN). An ITN is defined as a factory-treated net that does not require any further treatment.

Sample: Households

Full household ITN coverage

Percentage of households with at least one ITN for every two people.

Sample: Households

An ITN is defined as a factory-treated net that does not require any further treatment. In the 2012 and 2014 MMIS an ITN included nets that had been soaked with insecticides within the past 12 months. In the most recent questionnaire changes, The DHS Program dropped questions on retreatment of nets. This was done because bed nets that require annual retreatment and the products used for retreatment are no longer distributed and the distinction between ITNs and long-lasting insecticide-treated nets (LLIN) is no longer meaningful. For the 2021 MMIS, the current ITN was previously known as a LLIN in the 2014 and 2012 Malawi MIS surveys.

It is well understood that proper use of ITNs protects households and local communities from malaria. The distribution and use of ITNs is the central intervention for preventing malaria infection in Malawi. The 2017-2022 Malawi Malaria Control Strategic Plan prioritizes increasing household ownership of ITNs; from the 2014 baseline level of 70% of households with at least one ITN to cover 95% of households by the year 2022.

To protect people from malaria, the Malawi government, through the Malawi Ministry of Health (MoH) identified vector control and management as a critical intervention. One of the activities under this strategy is provision and promotion of the use of insecticide-treated nets (ITNs). In 2012, the National Malaria Control Programme (NMCP), in collaboration with partners, conducted the first nationwide mass distribution campaign, which was followed by three additional mass distribution campaigns between the years 2014 and 2016. For example, in 2014-16 a total of 8, 661, 990 ITNs were distributed to 12, 233, 548 people in 19 districts. In 2018, 10, 688, 831 ITNs were distributed to 17, 931, 638 people in 28 districts.

In addition to reaching all households across the country with ITN distribution, the national strategy aims to provide enough ITNs to cover all household residents. This indicator is operationalised as one ITN for every two household members

The 2021 MMIS revealed that 55% of households in Malawi own at least one ITN (**Table 3.1**). Twenty-one percent of households have at least one ITN for every two people who stayed in the household the night prior to the survey. It was revealed that a substantial number of households (45%) do not own any ITNs. In addition, the quantity of ITNs distributed needs to increase to provide sufficient ITNs for the 34% of households that own at least one ITN but have an insufficient supply for the number of household residents (**Figure 3.1**).

Figure 3.1 Household ownership of ITNs
Percent distribution of households

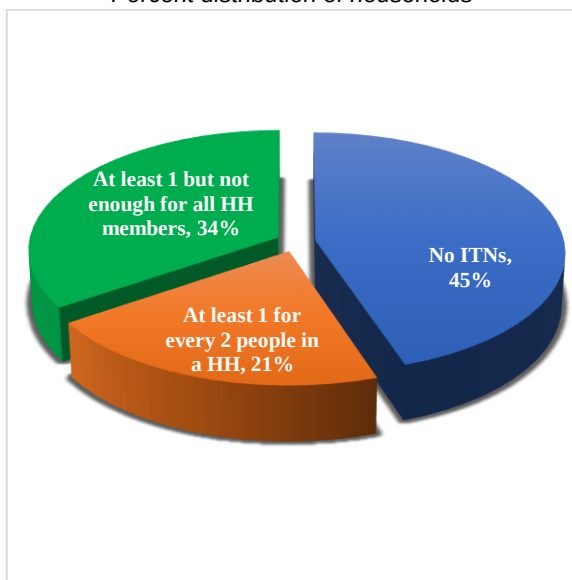
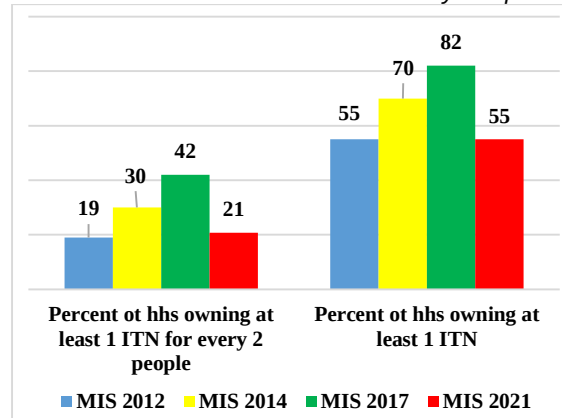


Figure 3.2 Percent distribution of ITN ownership over MIS years

Percentage of households owning at least one insecticide-treated net (ITN) and percentage of households with at least one net for every two persons



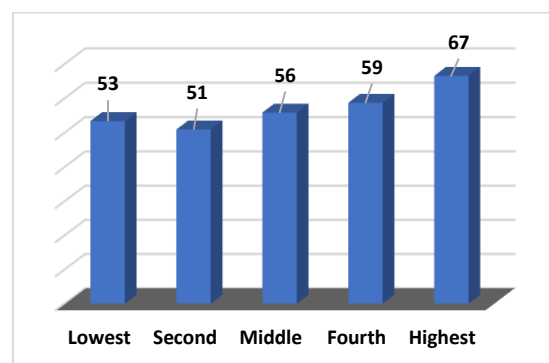
Patterns by background characteristics

- The percentage of households with at least one ITN increases as household wealth slightly
- Households in urban areas are more likely than households in rural areas to have at least one ITN for every two

increases from 53% in the lowest quintile to 67% in the highest wealth quintile (**Figure 3.3**).

Figure 3.3 ITN ownership by household wealth

Percentage of households with at least one ITN



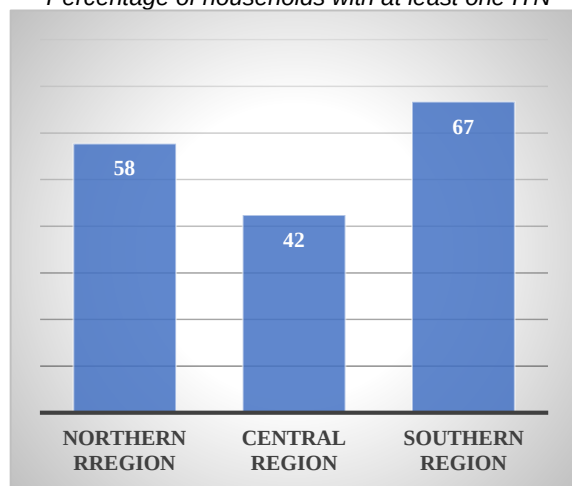
persons who stayed in the house the prior night (29% and 19%, respectively) (**Table 3.1**).

- The percentage of households with at least one ITN is highest in the Southern region

(67%) and lowest in the Central region (42%) (**Figure 3.4**).

Figure 3.4 ITN ownership by region

Percentage of households with at least one ITN

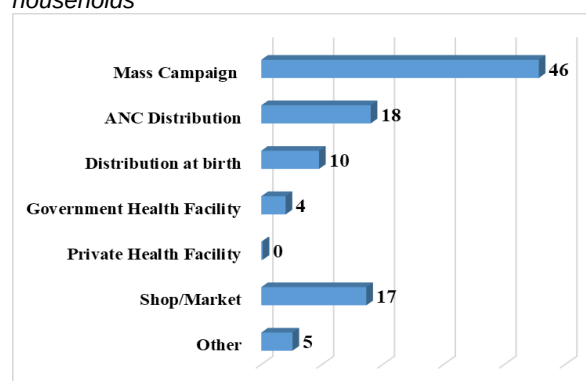


Source of Nets

The majority of ITNs (46%) were obtained from mass distribution campaigns. Another 18% of ITNs came from routine antenatal care (ANC) visits, 10% of ITNs were distributed to newborns at the time of birth, and 17% were purchased from shops or markets, and (**Figure 3.5** and **Table 3.2**).

Figure 3.5 Source of ITN

Percentage distribution of ITNs in interviewed households



3.2 HOUSEHOLD ACCESS AND USE OF ITNS

Access to an ITN

Percentage of the population that could sleep under an ITN if each ITN in the household were used by up to two people. **Sample:** De facto household population

Use of ITNs

Percentage of population that slept under an ITN the night before the survey.

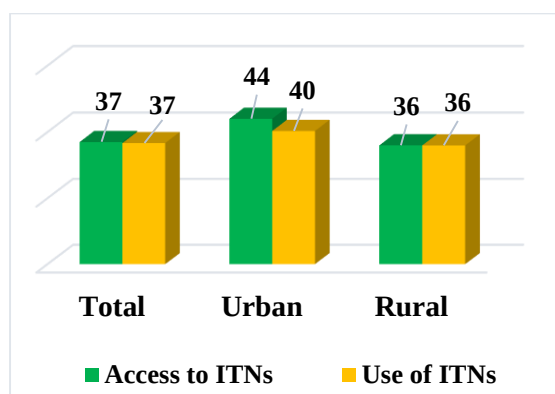
Sample: De facto household population

ITNs act as both a physical and a chemical barrier against mosquitoes. By reducing the vector population, ITNs may help to reduce malaria risk at the community level as well as to individuals who use them. Access to an ITN is measured by the proportion of the population that could sleep under an ITN if each ITN in the household were used by up to two people. Comparing ITN access and ITN use indicators can help programmes identify if there is a behavioural gap in which available ITNs are not being used. If the difference between these indicators is substantial, the programme may need to focus on behaviour change and how to identify the main drivers or barriers to ITN use to design an appropriate intervention. This analysis helps ITN programmes determine whether they need to achieve higher ITN coverage, promote

ITN use, or both. Thirty-seven percent of people in Malawi have access to an ITN, whereas 37% reported having slept under an ITN the night before the survey (**Table 3.4** and **Table 3.5**). Comparing these two population-level indicators, it is evident that they are close to each other at the population level. The gap between access and use of ITNs is slightly higher in urban (four percentage points) than in rural households that is almost the same (**Figure 3.6**).

Figure 3.6 Access to and use of ITN

Percentage of the household population with access to an ITN the night before the survey



Percentage of the household population that has access to an ITN and percentage of population that slept under an ITN the night before the survey

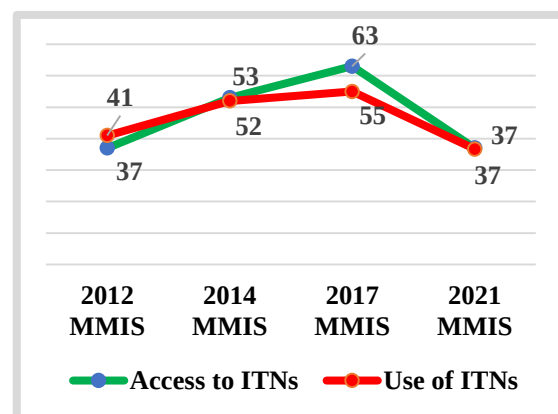


Figure 3.7 Trends in ITN access and use

Trends: The percentage of the household population with access to an ITN increased from 37% in the 2012 MMIS to 52% in the 2014 MMIS, 63% in the 2017 MMIS and decreased to 37% in the 2021 MMIS. The percentage of the household population that slept under an ITN the night before the survey increased from 41% in the 2012 MMIS to 53% in the 2014 MMIS, 55% in the 2017 MIS and decreased to 37% in 2021 MMIS. In both 2012 and 2014 MMIS surveys, there was a narrow gap between ITN access and ITN use while results from the 2017 MMIS show a wider gap. Contrary to 2021 MMIS that shows no between access and used (**Figure 3.7**).

In households with at least one ITN, only 64% of the household population slept under an ITN the previous night (**Table 3.5**). Overall, 83% of all existing ITNs were used the night before the survey (**Table 3.6**).

Patterns by background characteristics

- People in urban areas have greater access to ITNs in the household than those in rural areas (44% versus 36%) (**Table 3.4**).
- ITN access ranges from 27% in the Central region to 42% in Northern region and 46% in Southern regions (**Figure 3.8**).

3.3 USE OF ITNs BY CHILDREN AND PREGNANT WOMEN

Malaria is endemic in Malawi with transmission occurring throughout the year in most areas. The entire population of Malawi is at risk of malaria. Pregnant women, their unborn babies, and children under age 5 are at high risk of the negative consequences of malaria. Children under age 5 are

prone to severe malaria infection because they lack acquired immunity. For about 6 months after birth, antibodies acquired from the mother during pregnancy protect the child, although this maternal immunity is gradually lost when the child begins to develop his or her own immunity to malaria. Age is an important factor in determining levels of acquired immunity to malaria because acquired immunity does not prevent infection but protects against severe disease and death. The pace at which immunity develops depend on the exposure to malarial infection. In high malaria-endemic areas, children are thought to attain a high level of immunity by the fifth birthday. These children may experience an episode of malarial illness but usually do not suffer from severe, life-threatening conditions (Shulman and Dorman 2003).

Adults usually acquire some degree of immunity. However, since pregnancy suppresses immunity, women in their first pregnancies are at increased risk for severe malaria. Malaria in pregnancy is frequently associated with the development of anaemia, which interferes with the maternal-foetus exchange and may lead to low-birth-weight infants, placental parasitaemia, foetal death, abortion, stillbirth, and prematurity (Shulman and Dorman 2003).

The primary malaria prevention intervention in Malawi is the distribution of ITNs and promotion of ITN use. The Malawi Malaria Control Strategic Plan 2014-2020 emphasizes activities that promote the use of ITNs every night to prevent malaria complications. Strategies for ITN distribution in Malawi are (1) free routine distribution to pregnant women through ANC and to newborns at the time of delivery and (2) mass campaigns every two to three years.

level of immunity by the fifth birthday. These children may experience an episode of malarial illness but usually do not suffer from severe, life-threatening conditions (Shulman and Dorman 2003).

Figure 3.8 ITN access by region

Percent of the de facto population with access to an ITN

(MAP TO BE PROVIDED LATER)

Figure 3.9 ITN use

Percentage who slept under an ITN the night before the survey

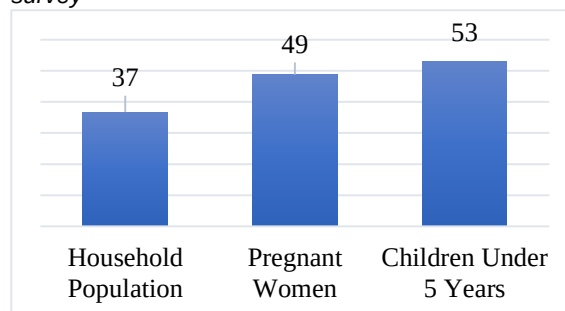


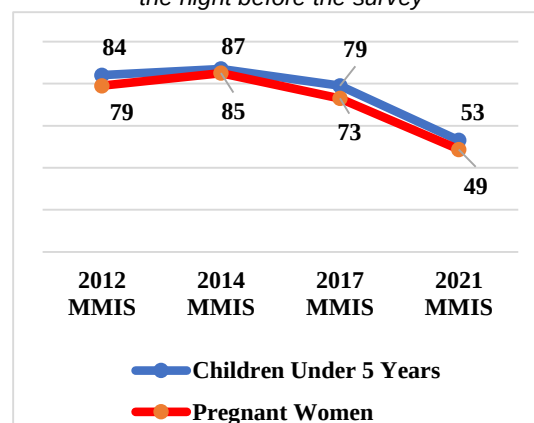
Table 3.7 and **Table 3.8** show the percentage of children under 5 and the percentage of pregnant women who slept under an ITN the night before the survey. Overall, 53% of children in Malawi under age 5 and 49% of pregnant women slept under an ITN the previous night (**Figure 3.9**).

In households with at least one ITN, 81% of children under age 5 and 77% of pregnant women slept under an ITN the night before the survey (**Table 3.7** and **Table 3.8**).

Trends: ITN use among children under age 5 in households with at least one ITN increased between 2012 and 2014 MMIS from 84% to 87%, however, it declined between 2014 and 2017 from 87% to 79% and kept on decreasing between 2017 and 2021, from 79% to 53%. As with children under age 5, the percentage of pregnant women in households with at least one ITN who slept under an ITN last night increased from 79% in the 2012 MMIS to 85% in the 2014 MMIS, but declined to 73% in the 2017 MMIS and kept on decreasing to 49% in the 2021 MMIS. (**Figure 3.10**).

Figure 3.10 Trends in ITN use by children and pregnant women

Among children under age 5 and pregnant women age 15-49 in households with at least one ITN, percentage who slept under an ITN the night before the survey



Patterns by background characteristics

- Children in urban areas are more likely than children in rural areas to sleep under an ITN among households with at least one ITN (85% and 80%, respectively) (**Table 3.7**).
- The percentage of pregnant women sleeping under an ITN the night before the survey is lowest among those in the Central region (68%) and highest in the Northern region (86%) in the households with at least one ITN (**Table 3.8**).

3.4 MALARIA IN PREGNANCY

Intermittent preventive treatment (IPTp) during pregnancy (IPTp2+)

Percentage of women who took at least two doses of SP/Fansidar during their last pregnancy.

Sample: Women age 15-49 with a live birth in the 2 years before the survey

Intermittent preventive treatment (IPTp) during pregnancy (IPTp3+)

Percentage of women who took at least three doses of SP/Fansidar during their last pregnancy.

Sample: Women age 15-49 with a live birth in the 2 years before the survey

Malaria infection during pregnancy is a major public health problem in Malawi, with substantial risks for the mother, her foetus, and the neonate. Intermittent preventive treatment of malaria in pregnancy (IPTp) is a full therapeutic course of antimalarial medicine given to pregnant women at routine antenatal care visits to prevent malaria. IPTp helps prevent maternal malaria episodes, maternal and foetal anaemia, placental parasitaemia, low birth weight, and neonatal mortality.

The World Health Organization (WHO) recommends a three-pronged approach for reducing the negative health effects associated with malaria in pregnancy: prompt diagnosis and treatment of confirmed infection, use ITNs, and IPTp (WHO 2004).

Sulfadoxine-pyrimethamine (SP), also known as Fansidar, is the recommended drug for IPTp in Malawi. For over 10 years, the Malawi Ministry of Health (MoH) has been implementing IPTp, defined as provision of at least two doses of sulfadoxine-pyrimethamine (SP)/Fansidar to protect the mother and her child from malaria during routine antenatal care visits in the second and third trimesters of pregnancy (IPTp2+). In 2016 the National Malaria Control Programme adopted the 2012 WHO recommendation to administer one dose of SP/Fansidar at each antenatal care (ANC) visit after the first trimester, with at least 1 month between doses (WHO 2012a; WHO 2012b). The household survey indicator used to measure coverage of this intervention is the percentage of women with a live birth in the 2 years preceding the survey who received three or more doses of SP/Fansidar to prevent malaria during her most recent

pregnancy (IPTp3+).

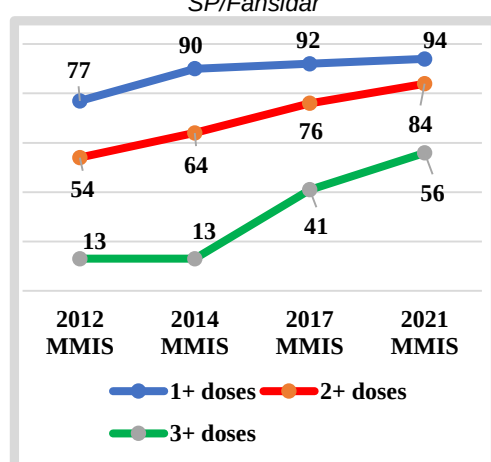
Ninety-four percent of women with a live birth in the 2 years preceding the survey

received one or more doses of SP/Fansidar to prevent malaria. Eighty-four percent of the women received two or more doses of SP/Fansidar, and 56% of the women received three or more doses of SP/Fansidar. (Table 3.9).

Trends: The percentage of women receiving IPTp1+ increased from 77% in 2012 MMIS to 90% in the 2014 MMIS, 92% in the 2017 MMIS and to 94% in the current survey. The percentage of women receiving two or more doses of SP/Fansidar for IPTp increased from 54% in in 2012 MMIS to 64% in the 2014 MMIS, 76% in 2017 and to 84% in the 2021 MMIS. There was an increase in IPTp3+ between the 2014 MMIS (13%) to 41% in 2017 and increased further to 56% in 2021 MMIS (Figure 3.11).

Figure 3.11 Trends in IPTp use by pregnant women

Percentage of women with a live birth in the 2 years before the survey who received at least 1, 2, or 3 doses of SP/Fansidar



Patterns by background characteristics

- The use of IPTp1+ (93% and 95%), IPTp2+ (83% and 84%) and use of IPTp3+ is similar (56%) among women in the urban and rural areas respectively (**Table 3.9**).
- IPTp3+ coverage ranges from 49% in the Southern region to 64% in the Central region.

3.5 MOSQUITO NET PREFERENCE

Preferences for various social marketing goods significantly affect the consistent use of products. In consideration of this influence, the 2021 MMIS observed actual colour of respondents' mosquito nets as well as assessed respondents' preferences for colour. Additionally, the survey also observed the shape of mosquito nets and assessed respondents' preferred shape. In recent years, nets obtained through the public sector (mass distribution campaigns and/or routine distribution) have been green, while nets obtained through the private sector (shops or markets) are predominantly blue or white.

In assessing the colour of the nets, 69% of the observed nets were green, 17% white, 8.5% light blue and 5% dark blue (**Table 3.10**). However, when the respondents were asked about colour preference, 45% preferred the blue colour, and 35% preferred the green (**Table 3.11**).

In assessing the shape of the nets, 88% of the observed nets were rectangular and 12% were conical (**Table 3.12**). However, 68% of respondents prefer conical, 30% prefer rectangular, and 1% did not have a clear preference (**Table 3.13**).

Among households that preferred conical nets, 84% said that they are easier to hang, 26% said that they fit better around sleeping spaces and 20% said they are stronger. Similarly, among households that preferred rectangular nets, 63% said that these nets are easier to hang, 33% said that they are a better fit around sleeping spaces and 23% said they look nice (**Table 3.14**).

Trends: Preferences in colour have changed from 38% green and 36% blue in the 2012 MMIS to 29% green and 61% blue in the 2017 MMIS and then to 35% green and 45% blue in the 2021 MMIS (**Figure 3.12**).

Additionally, preferences in shape have changed from 76% conical and 21% rectangular in the 2017 MMIS to 68% conical and 30% rectangular in the 2021 MMIS (**Figure 3.13**).

Figure 3.12 Trends in colour preference

Percentage of households by preference of colour of mosquito net

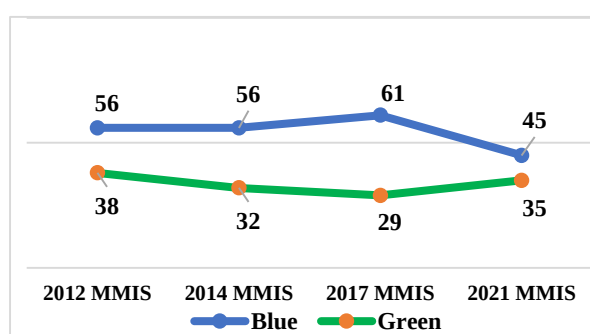
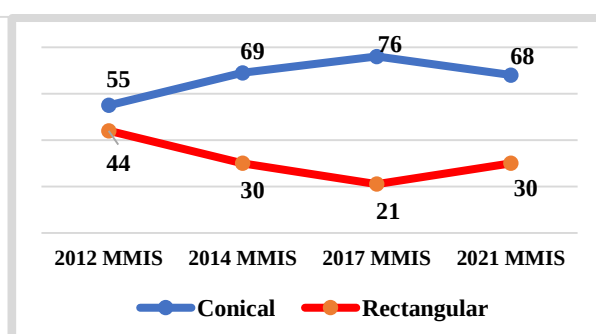


Figure 3.13 Trends in preferred net

Percentage of households by preference of shape of mosquito net



LIST OF TABLES

For detailed information on malaria, see the following tables:

- **Table 3.1 Household possession of mosquito nets**
- **Table 3.2 Source of mosquito nets**
- **Table 3.3 Access to an insecticide-treated net (ITN)**
- **Table 3.4 Access to an insecticide-treated net (ITN) by background characteristics**
- **Table 3.5 Use of mosquito nets by persons in the household**
- **Table 3.6 Use of existing ITNs**
- **Table 3.7 Use of mosquito nets by children**
- **Table 3.8 Use of mosquito nets by pregnant women**
- **Table 3.9 Use of intermittent preventive treatment (IPTp) by women during pregnancy**
- **Table 3.10 Observed colour of mosquito nets**
- **Table 3.11 Preferred colour of mosquito nets**
- **Table 3.12 Observed shape of mosquito nets**
- **Table 3.13 Preferred shape of mosquito nets**
- **Table 3.14 Reasons for preferring a specific shape of mosquito net**

Table 3.1 Household possession of mosquito nets

Percentage of households with at least one mosquito net (treated or untreated) and insecticide-treated net (ITN); average number of nets and ITNs per household; and percentage of households with at least one net and ITN per two persons who stayed in the household last night, according to background characteristics, Malawi MIS 2021

Malawi MRS 2021								Numb er of house holds with at least one person who stayed in the house hold last night
Background characteristic	Percentage of households with at least one mosquito net		Average number of nets per household		Numb er of house -holds	Percentage of households with at least one mosquito net for every two persons who stayed in the household last night		
	Any mosqu ito net	Insecticide- treated mosquito net (ITN) ¹	Any mosquit o net	Inse ctici de- treat ed mos quito net (ITN) ¹		Any mosquit o net	Insecticide -treated mosquito net (ITN) ¹	
Residence								
Urban	67.2	55.9	1.4	1.1	607	36.5	28.7	605
Rural	59.0	55.0	0.9	0.9	3,104	21.2	19.2	3,094
Region								
Northern Region	62.8	57.7	1.2	1.1	420	28.5	25.0	418
Central Region	48.5	42.4	0.8	0.6	1,604	17.0	13.6	1,601
Southern Region	71.0	66.7	1.2	1.1	1,687	28.9	26.5	1,680
Wealth quintile								
Lowest	55.3	53.4	0.8	0.7	1,187	16.3	14.9	1,178

Second	56.7	50.9	0.8	0.7	978	19.1	16.9	976
Middle	60.1	55.9	1.1	1.0	685	23.0	21.0	684
Fourth	65.3	58.7	1.2	1.1	511	30.5	27.7	511
Highest	80.5	66.5	2.1	1.6	350	53.1	40.3	350
Total	60.3	55.2	1.0	0.9	3,711	23.7	20.7	3,699

¹ An insecticide-treated net (ITN) is a factory-treated net that does not require any further treatment. In the MMIS 2017, this was known as a long-lasting insecticidal net (LLIN).

Note to survey managers: the second sentence in the footnote should reference the most recent, prior DHS. So, for example, if the current X DHS was conducted in 2017 and the most recent prior one was in 2012, the second sentence of the footnote should read "In the 2012 X DHS, this was known as a long-lasting insecticidal net (LLIN)."

Table 3.2 Source of mosquito nets

Percent distribution of mosquito nets by source of net, according to background characteristics, Malawi MIS 2021

Background characteristic	Mass distribution campaign	ANC visit	Immunitization visit	Government health facility	Private health facility	Pharmacy	Shop/market	Community health worker	Religious institution	School	CHAM/Mission	Other	Don't know	Total	Number of mosquito nets
Type of net															
ITN ¹	49.8	20.3	10.0	0.7	3.6	0.0	0.1	0.2	7.5	1.7	0.3	5.4	0.3	100.0	3,310
Other ²	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	100.0	0.0	0.0	0.0	0.0	100.0	40
Residence															
Urban	39.4	8.7	5.0	0.5	4.9	0.0	0.1	0.4	35.3	1.5	0.3	3.7	0.1	100.0	84
Rural	45.9	20.9	10.0	0.7	2.8	0.0	0.0	0.1	12.4	1.5	0.2	5.1	0.4	100.0	2,880
Region															
Northern Region	52.8	12.6	7.1	0.3	5.4	0.0	0.2	0.1	17.4	1.1	0.0	3.0	0.1	100.0	52
Central Region	34.2	18.4	10.5	0.4	5.3	0.0	0.0	0.2	23.0	2.8	0.4	4.3	0.3	100.0	1,230
Southern Region	48.5	19.4	8.4	0.9	1.4	0.0	0.1	0.1	14.3	0.8	0.1	5.6	0.4	100.0	1,960
Wealth quintile															
Lowest	46.1	27.3	13.9	0.1	1.3	0.0	0.0	0.0	7.1	0.2	0.0	3.8	0.3	100.0	90
Second	39.1	21.9	12.9	0.8	3.1	0.0	0.1	0.0	14.7	2.3	0.2	4.8	0.2	100.0	77
Middle	50.6	18.2	7.0	1.4	2.3	0.0	0.1	0.0	11.2	1.4	0.0	7.2	0.5	100.0	72
Fourth	47.3	13.4	5.7	0.6	4.2	0.0	0.0	0.0	20.0	2.7	0.0	5.4	0.7	100.0	60
Highest	39.2	6.3	2.9	0.5	6.0	0.0	0.0	0.8	38.3	1.5	1.0	3.3	0.0	100.0	72
Total	44.4	18.1	8.9	0.7	3.2	0.0	0.1	0.2	17.6	1.5	0.2	4.8	0.3	100.0	3,720

ANC = Antenatal care

¹ An insecticide-treated net (ITN) is a factory-treated net that does not require any further treatment. In the MMIS 2017, this was known as a long-lasting insecticidal net (LLIN).

² Any net that is not an ITN

Table 3.3 Access to an insecticide-treated net (ITN)

Percent distribution of the de facto household population by number of ITNs the household owns, and percentage with access to an ITN, according to number of persons who stayed in the household the night before the survey, Malawi MIS 2021

	Number of persons who stayed in the household the night before the survey								
Number of ITNs ¹	1	2	3	4	5	6	7	8+	Total
0	62.6	49.5	44.1	41.4	42.9	38.9	38.7	47.2	42.9
1	32.6	38.2	40.2	35.3	25.9	29.7	24.5	17.3	29.7
2	3.6	10.2	11.7	17.6	19.8	15.0	23.6	16.2	16.6
3	0.4	2.0	3.5	3.9	8.6	12.9	8.9	9.1	7.3
4	0.9	0.0	0.5	1.4	2.5	2.2	3.6	7.7	2.6
5	0.0	0.1	0.0	0.3	0.3	0.4	0.5	1.3	0.4
6	0.0	0.0	0.1	0.1	0.1	0.4	0.0	0.6	0.2
7	0.0	0.0	0.0	0.0	0.0	0.5	0.1	0.7	0.2
8+									
Total	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
Number	277	854	2,203	2,843	2,916	2,678	1,967	2,218	15,956
Percentage with access to an ITN ^{1,2}	37.4	50.55	42.5	40.9	37.7	36.4	32.4	26.9	37.4

¹ An insecticide-treated net (ITN) is a factory-treated net that does not require any further treatment. In the MMIS 2017, this was known as a long-lasting insecticidal net (LLIN).

² Percentage of the de facto household population who could sleep under an ITN if each ITN in the household were used by up to two people

Table 3.4 Access to an ITN

Percentage of the de facto population with access to an ITN in the household, according to background characteristics, Malawi MIS 2021

Background characteristic	Percentage of the de facto population with access to an ITN ¹
Residence	
Urban	43.8
Rural	35.1
Region	
Northern Region	41.8
Central Region	26.8
Southern Region	46.3
Wealth quintile	
Lowest	33.9
Second	31.6
Middle	38.4
Fourth	41.6
Highest	53.5
Total	37.4

¹ Percentage of de facto household population who could sleep under an ITN if each ITN in the household were used by up to two people

Table 3.5 Use of mosquito nets by persons in the household

Percentage of the de facto household population who slept the night before the survey under a mosquito net (treated or untreated) and under an insecticide-treated net (ITN); and among the de facto household population in households with at least one ITN, percentage who slept under an ITN the night before the survey, according to background characteristics, Malawi MIS 2021

Background characteristic	Household population			Household population in households with at least one ITN ¹	
	Percentage who slept under any mosquito net last night	Percentage who slept under an ITN ¹ last night	Number of persons	Percentage who slept under an ITN ¹ last night	Number of persons
Age					
<5	58.1	53.3	2,062	81.3	1,352
5-14	31.5	28.3	4,861	50.7	2,711
15-34	40.6	36.6	5,245	64.1	2,995
35-49	47.6	41.8	1,890	72.8	1,086
50+	39.6	35.1	1,625	71.0	804
Sex					
Male	38.5	34.6	7,589	61.8	4,247
Female	43.0	38.6	8,095	66.6	4,701
Residence					
Urban	50.5	40.3	2,518	69.4	1,464
Rural	39.0	36.0	13,166	63.3	7,484
Region					
Northern Region	40.5	35.5	1,901	59.9	1,128
Central Region	32.3	27.7	6,742	63.1	2,964
Southern Region	49.1	45.5	7,042	66.0	4,856
Wealth quintile					
Lowest	37.8	36.2	4,660	63.8	2,642
Second	36.4	32.8	4,017	62.8	2,100
Middle	41.1	37.8	3,110	66.4	1,772
Fourth	41.0	36.3	2,278	62.4	1,327
Highest	59.6	46.1	1,620	67.5	1,107
Total	40.8	36.7	15,684	64.3	8,948

¹ An insecticide-treated net (ITN) is a factory-treated net that does not require any further treatment. In the MMIS 2017, this was known as a long-lasting insecticidal net (LLIN).

Table 3.6 Use of existing ITNs

Percentage of insecticide-treated nets (ITNs) that were used by anyone the night before the survey, according to background characteristics, Malawi MIS 2021

Background characteristic	Percentage of existing ITNs ¹ used last night	Number of ITNs ¹
Residence		
Urban	84.9	668
Rural	82.7	2,648
Region		
Northern Region	81.7	465
Central Region	83.3	1,033
Southern Region	83.4	1,817
Wealth quintile		
Lowest	81.2	860
Second	84.9	690
Middle	85.6	666
Fourth	82.2	537
Highest	81.9	563
Total	83.1	3,316

¹ An insecticide-treated net (ITN) is a factory-treated net that does not require any further treatment. In the MMIS 2017, this was known as a long-lasting insecticidal net (LLIN).

Table 3.7 Use of mosquito nets by children

Percentage of children under age 5 who, the night before the survey, slept under a mosquito net (treated or untreated) and under an insecticide-treated net (ITN); and among children under age 5 in households with at least one ITN, percentage who slept under an ITN the night before the survey, according to background characteristics, Malawi MIS 2021

Background characteristic	Children under age 5 in all households			Children under age 5 in households with at least one ITN ¹	
	Percentage who slept under any mosquito net last night	Percentage who slept under an ITN ¹ last night	Number of children	Percentage who slept under an ITN ¹ last night	Number of children
Age in months					
<12	73.1	68.5	378	86.8	298
12-23	64.3	59.5	453	83.7	322
24-35	53.4	47.8	450	79.8	269
36-47	52.7	47.3	426	81.0	249
48-59	44.9	42.0	368	70.7	218
Sex					
Male	57.6	52.9	1,011	81.6	655
Female	57.8	53.1	1,064	80.6	702
Residence					
Urban	65.5	54.8	283	85.2	182
Rural	56.5	52.7	1,792	80.4	1,175
Region					
Northern Region	55.4	49.4	217	77.2	139
Central Region	46.3	41.6	878	80.3	455
Southern Region	68.5	64.0	980	82.2	763

Wealth quintile

Lowest	53.6	51.4	704	78.3	462
Second	57.1	52.0	600	82.8	376
Middle	61.1	56.5	370	84.7	247
Fourth	56.7	51.6	244	79.3	159
Highest	72.6	58.0	157	81.2	112
Total	57.7	53.0	2,075	81.1	1,357

Note: Table is based on children who stayed in the household the night before the interview.

¹ An insecticide-treated net (ITN) is a factory-treated net that does not require any further treatment. In the MMIS 2017, this was known as a long-lasting insecticidal net (LLIN).

Table 3.8 Use of mosquito nets by pregnant women

Percentage of pregnant women age 15-49 who, the night before the survey, slept under a mosquito net (treated or untreated) and under an insecticide-treated net (ITN); and among pregnant women age 15-49 in households with at least one ITN, percentage who slept under an ITN the night before the survey, according to background characteristics, Malawi MIS 2021

Background characteristic	Among pregnant women age 15-49 in all households			Among pregnant women age 15-49 in households with at least one ITN ¹	
	Percentage who slept under any mosquito net last night	Percentage who slept under an ITN ¹ last night	Number of pregnant women	Percentage who slept under an ITN ¹ last night	Number of pregnant women
Residence					
Urban	51.7	45.6	40	71.4	26
Rural	52.7	49.3	222	78.1	140
Region					
Northern Region	66.0	55.8	30	86.2	20
Central Region	36.2	30.9	109	67.9	50
Southern Region	63.7	62.9	123	79.9	97
Education					
No education	27.1	27.1	14	59.9	6
Primary	52.1	50.1	172	78.5	110
Secondary	57.9	49.9	73	75.4	48
More than secondary	61.4	42.5	4	100.0	2
Wealth quintile					
Lowest	51.0	48.4	83	71.1	57
Second	48.0	40.7	60	85.4	29
Middle	59.3	58.2	65	92.0	41
Fourth	56.7	53.0	29	80.0	19
Highest	45.8	39.4	25	48.7	20
Total	52.5	48.7	263	77.1	166

Note: Table is based on women who stayed in the household the night before the interview.

¹ An insecticide-treated net (ITN) is a factory-treated net that does not require any further treatment. In the MMIS 2017, this was known as a long-lasting insecticidal net (LLIN).

Table 3.9 Use of intermittent preventive treatment (IPTp) by women during pregnancy

Percentage of women age 15-49 with a live birth in the 2 years preceding the survey who, during the pregnancy that resulted in the last live birth, received one or more doses of SP/Fansidar, received two or more doses of SP/Fansidar, and received three or more doses of SP/Fansidar, according to background characteristics, Malawi MIS 2021

Background characteristic	Percentage who received one or more doses of SP/Fansidar ¹	Percentage who received two or more doses of SP/Fansidar	Percentage who received three or more doses of SP/Fansidar	Number of women with a live birth in the 2 years preceding the survey
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Residence				
Urban	93.4	82.9	56.0	117
Rural	94.6	84.1	56.0	755
Region				
Northern Region	94.8	83.3	59.3	86
Central Region	96.6	89.2	63.8	367
Southern Region	92.5	79.5	48.5	419
Education				
No education	89.9	82.6	56.9	92
Primary	95.5	83.9	54.8	579
Secondary	93.6	85.6	58.7	186
More than secondary	92.4	74.4	62.0	14
Wealth quintile				
Lowest	96.8	85.0	54.9	311
Second	93.4	87.4	59.0	240
Middle	94.8	81.3	59.1	161
Fourth	90.4	78.4	48.4	100
Highest	92.6	80.9	54.0	59
Total	94.4	83.9	56.0	872

Note to survey managers: whether or not at least one dose of SP/Fansidar was received during an ANC visit is no longer part of the IPTp indicator.

Table 3.10 Observed colour of mosquito nets
Percent distribution of mosquito nets by observed colour of mosquito net, according to background characteristics, Malawi MIS, 2021

Background characteristic		Observed colour of mosquito net								Number of mosquito nets
		Green	Dark blue	Light blue	Red	Black	White	Other	Total	
residence	Urban	61.9	7.2	12.0	0.1	0.0	16.7	2.0	100.0	842
	Rural	70.8	3.8	7.4	0.1	0.0	17.3	0.6	100.0	2,880
region	Northern Region	73.0	6.8	3.8	0.1	0.1	14.9	1.3	100.0	525
	Central Region	62.6	3.5	12.7	0.2	0.0	19.5	1.5	100.0	1,237
	Southern Region	71.5	4.6	7.0	0.1	0.0	16.4	0.4	100.0	1,961
Wealth quintile	Lowest	68.0	2.6	7.8	0.0	0.0	21.2	0.3	100.0	904
	Second	71.8	2.7	8.1	0.2	0.0	16.5	0.8	100.0	772
	Middle	76.1	3.8	4.8	0.2	0.1	14.5	0.5	100.0	726
	Fourth	70.0	5.9	7.2	0.0	0.0	16.2	0.7	100.0	600
	Highest	57.9	8.6	14.4	0.1	0.0	16.5	2.5	100.0	720
Total		68.7	4.6	8.5	0.1	0.0	17.2	0.9	100.0	3,722

Table 3.11 Preferred colour of mosquito nets
Percent distribution of households by preferred colour of mosquito net, according to background characteristic, Malawi MIS 2017

Background characteristic		Preferred colour of mosquito nets								Number of households
		Blue	Green	Red	White	Black	Other	Dont know / no preference	Total	
residence	Urban	52.6	22.7	0.4	20.4	0.7	1.4	1.7	100.0	607
	Rural	44.0	36.9	0.7	14.7	0.2	0.8	2.7	100.0	3,104
region	Northern Region	50.0	30.0	0.8	16.0	0.3	1.0	1.9	100.0	420
	Central Region	42.6	36.6	0.9	14.9	0.2	1.1	3.7	100.0	1,604
	Southern Region	46.9	33.9	0.3	16.2	0.4	0.6	1.7	100.0	1,687
Wealth quintile	Lowest	37.6	42.7	1.1	14.7	0.4	0.6	2.9	100.0	1,187
	Second	45.3	36.7	0.5	12.4	0.2	1.0	3.8	100.0	978
	Middle	47.4	32.5	0.5	17.1	0.6	0.6	1.4	100.0	685
	Fourth	52.7	27.6	0.2	17.0	0.0	1.1	1.4	100.0	511
	Highest	57.5	15.7	0.3	22.9	0.3	1.4	1.9	100.0	350
Total		45.4	34.6	0.6	15.6	0.3	0.9	2.6	100.0	3,711

Table 3.12 Observed shape of mosquito nets
Percent distribution of mosquito nets by observed shape of mosquito net, according to background characteristics, Malawi MIS, 2021

Background characteristic	Observed shape of mosquito net				Number of mosquito nets
	Conical	Rectangular	Other	Total	

residence	Urban	29.0	71.0	0.0	100.0	842
	Rural	7.1	92.8	0.1	100.0	2,880
region	Northern Region	11.7	88.2	0.1	100.0	525
	Central Region	16.4	83.6	0.0	100.0	1,237
	Southern Region	9.4	90.5	0.2	100.0	1,961
Wealth quintile	Lowest	4.0	95.7	0.3	100.0	904
	Second	5.4	94.6	0.1	100.0	772
	Middle	6.0	94.0	0.0	100.0	726
	Fourth	12.9	87.1	0.0	100.0	600
	Highest	34.6	65.4	0.0	100.0	720
	Total	12.0	87.9	0.1	100.0	3,722

Table 3.13 Preferred shape of mosquito nets
Percent distribution of households by preferred shape of mosquito net,
by background characteristic
Malawi MIS, 2021

Background characteristic		Preferred shape of mosquito nets				Number of households
		Conical	Rectangular	Don't know / no preference	Total	
residence	Urban	83.9	15.0	1.1	100.0	607
	Rural	65.3	33.2	1.5	100.0	3,104
region	Northern Region	67.5	31.0	1.6	100.0	420
	Central Region	63.2	34.1	2.6	100.0	1,604
	Southern Region	73.4	26.4	0.2	100.0	1,687
Wealth quintile	Lowest	55.4	42.6	2.0	100.0	1,187
	Second	66.7	31.9	1.4	100.0	978
	Middle	73.1	25.3	1.6	100.0	685
	Fourth	80.9	18.4	0.7	100.0	511

Highest	88.6	11.0	0.4	100.0	350
Total	68.3	30.3	1.4	100.0	3,711

Table 3.14 Reasons for preferring a specific shape of mosquito nets, percentage that reported specific reasons for preferring the canonical shape, according to Malawi MIS, 2021

		Among households that preferred canonical mosquito nets, percentage that reported that it is because									Among households that preferred alternative mosquito nets, percentage that reported that it is because				
		they are easier to hang	they are easier to store when not hung	they are easier to travel with outside the household	they are better fit around sleeping place	more people can sleep under net (wider)	they look nicer	they are stronger	there are other reasons	Number of households that preferred mosquito nets in canonical shape	they are easier to hang	they are easier to store when not hung	they are easier to travel with outside the household	they are better fit around sleeping place	more people can sleep under net (wider)
Background characteristic	residence														
	Urban	87.3	3.8	0.9	26.0	7.5	7.5	18.0	9.2	510	52.2	2.8	1.4	41.6	1.0
	Rural	82.7	2.3	1.5	26.3	9.0	10.2	20.7	8.6	2,025	64.0	1.5	1.5	32.1	3.0
region	Northern Region	89.5	4.9	2.7	27.5	5.7	10.6	17.1	4.1	283	56.4	2.7	1.4	41.7	1.0
	Central Region	87.4	2.0	1.5	23.5	9.5	6.2	17.1	9.8	1,014	69.0	1.1	0.8	34.3	5.0
	Southern Region	79.1	2.6	1.0	28.3	8.7	12.3	23.4	8.9	1,237	57.6	2.0	2.3	28.6	1.0
Wealth quintile	Lowest	80.0	1.5	1.6	22.4	6.7	9.6	21.6	8.4	658	66.6	0.9	1.7	28.4	3.0
	Second	82.1	2.7	1.6	26.5	9.0	10.6	19.3	10.3	652	60.8	2.1	1.1	32.5	3.0
	Middle	81.6	4.1	0.8	29.7	11.9	12.6	22.0	7.6	501	62.3	2.5	1.6	35.8	3.0
	Fourth	88.1	2.4	1.9	27.8	9.3	6.1	16.9	8.6	413	55.4	1.8	1.5	46.4	4.0
	Highest	91.8	2.9	0.6	26.1	6.1	8.1	20.4	8.0	310	57.0	3.8	0.0	48.7	0.0
	Total	83.6	2.6	1.4	26.3	8.7	9.7	20.1	8.7	2,535	63.0	1.6	1.5	32.9	3.0

MALARIA IN CHILDREN

4

Key Findings

- **Fever prevalence:** Thirty-seven percent of children under age 5 had a fever in the 2 weeks prior to the survey.

- **Care-seeking for fever:** Advice or treatment was sought for 93% of children under age 5 with fever in the 2 weeks before the survey.
- **Source of advice or treatment:** Among children under age 5 with fever for whom advice or treatment was sought, 93% were from the public health sector, 7% were from private sector sources.
- **Testing:** Forty-one percent of children under age 5 with a fever had blood taken from a finger or heel for testing.
- **Type of antimalarial drug used:** Thirty-two percent of children under age 5 with fever who took an antimalarial drug in the 2 weeks before the survey were given an Artemisinin-based combination therapy (ACT), the recommended first-line malaria treatment in Malawi.
- **Severe anaemia:** One percent of children age 6-59 months were severely anaemic (haemoglobin level less than 8 g/dl).
- **Malaria:** Analysis of blood smears by microscopy revealed that 10.5% of children age 6-59 months had malaria parasites.

This chapter presents data useful for assessing how well fever management strategies are implemented. Specific topics include care seeking for febrile children, diagnostic testing of children with fever, and therapeutic use of antimalarial drugs. Prevalence of anaemia and malaria among children age 6-59 months is also assessed.

4.1 CARE SEEKING FOR FEVER IN CHILDREN

Care seeking for children under age 5 with fever

Percentage of children under age 5 with a fever in the 2 weeks before the survey for whom advice or treatment was sought from the public health sector, CHAM/mission, private medical sector, shop, market, and itinerant drug seller.

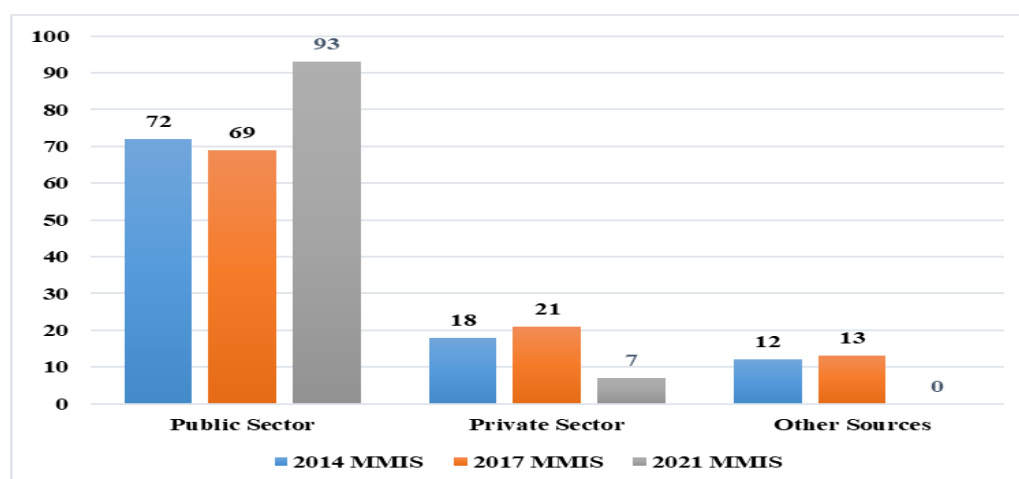
Sample: Children under age 5 with a fever in the 2 weeks before the survey

One of the key case management objectives of the National Malaria Control Programme (NMCP) is to ensure that all suspected malaria cases have access to confirmatory diagnosis and receive effective treatment. Fever is a key symptom of malaria and other acute infections in children. These fevers require prompt and effective treatment to prevent morbidity and mortality. Thirty-seven percent of children under age 5 had fever in the 2 weeks preceding the survey. Advice or treatment was sought for 93% of the children with fever in the 2 weeks preceding the survey, and timely care seeking (the same or next day following fever onset) occurred for 46% of the febrile children (Table 4.1).

Among the children with recent fever for whom care was sought, most received advice or treatment from the public health sector (93%), including 39% who sought care from a government health centre, and 30% from a government hospital. Only 7% sought advice from a private health sector (Table 4.2).

Figure 4.1 Trends in care seeking for fever in children by source of care

Among children under age 5 with fever for whom advice or treatment was sought, percentage for whom advice or treatment was sought from specific sources



Trends: Among those with fever, advice or treatment seeking was less likely in the 2021 MMIS than in the 2017 MMIS (54% compared to almost none). While the percentage of children with fever in the 2 weeks before the survey for whom advice or treatment was sought from the public sector increased between 2017 and 2021 (from 69% to 93%), the percentage of those for whom advice or treatment was sought in the private sector decreased from 21% to 7% within the same period (**Figure 4.1**).

Patterns by background characteristics

- The percentage of children under age 5 with fever decreases with increasing wealth quintile, dropping from 39% among children in the lowest wealth quintile to 24% among children in the highest wealth quintile.
- The percentage of children under age 5 with fever ranges from 26% in urban areas to 39% in rural areas.
- Advice or treatment for children with fever was sought for 90% of children age <12 months compared with 95% of children age 48-59 months.
- The percentage of children with fever for whom advice or treatment was sought ranges from 93% in the Northern region to 94% in the Southern region.

4.2 DIAGNOSTIC TESTING OF CHILDREN WITH FEVER

Diagnosis of malaria in children under 5 with fever

Percentage of children under 5 with a fever in the 2 weeks before the survey who had blood taken from a finger or heel for testing. This is a proxy measure of diagnostic testing for malaria.

Sample: Children under 5 with a fever in the 2 weeks before the survey

The Malawi National Malaria Control Programme policy recommends prompt parasitological confirmation before treatment commences. Malaria rapid diagnostic tests (RDTs) are performed for all patients suspected of uncomplicated malaria. Adherence to this policy cannot be directly measured through household surveys; however, the 2021 MMIS asked interviewed women with children under 5 who had a fever in the 2 weeks before the survey if the child had blood taken from a finger or heel for testing during the illness. This information is used as a proxy measure for adherence to the NMCP treatment guidelines of conducting diagnostic testing for all suspected malaria cases.

In the MMIS 2021, 41% of children under age 5 with a fever in the 2 weeks before the survey had blood taken from a finger or heel, presumably for malaria testing (**Table 4.1**).

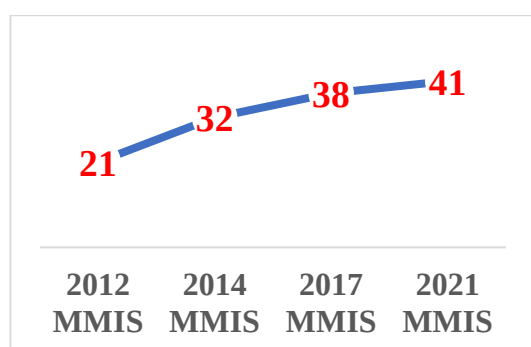
Trends: The percentage of children under age 5 with fever who had blood taken from the finger or heel for testing increased from 21% in the 2012 MMIS to 32% in 2014, 38% in the 2017% and then to 41% in the 2021 MMIS (**Figure 4.2**).

Patterns by background characteristics

- Forty-five percent of children less than age 12-23 months had blood taken from a finger or heel for testing, compared with 35% of children age 48-59 months.
- Forty-three percent of children under age 5 with recent fever in the Southern region had blood taken from a finger or heel for testing, compared with 53% from the Northern region (**Table 4.1**).

Figure 4.2 Trends in diagnostic testing of children with fever

Percent of children under age 5 with fever in the 2 weeks preceding the survey who had blood taken from a finger or heel for testing



4.3 USE OF RECOMMENDED ANTIMALARIALS

Artemisinin-based combination therapy (ACT) for children under 5 with fever

Among children under age 5 with a fever in the 2 weeks before the survey who took any antimalarial drugs, the percentage who took an artemisinin-based combination therapy (ACT)

Sample: Children under 5 with a fever in the two weeks before the survey

Artemisinin-based combination therapy (ACT) is the recommended first-line antimalarial drug for the treatment of uncomplicated malaria in Malawi. This policy was first recommended in 2006 and then implemented in 2007 (MoH 2013).

According to the treatment data in **Table 4.3**, 89.2% of children under age 5 with recent fever who received an antimalarial took an ACT.

4.4 PREVALENCE OF LOW HAEMOGLOBIN IN CHILDREN

Prevalence of low haemoglobin in children

Percentage of children age 6-59 months who had a haemoglobin measurement of less than 8 grams per decilitre (g/dl) of blood. The cut-off of 8 g/dl is often used to classify malaria-related anaemia. **Sample:** Children age 6-59 months

Anaemia, defined as a reduced level of haemoglobin in blood, decreases the amount of oxygen reaching the tissues and organs of the body and reduces their capacity to function. Anaemia is associated with impaired motor and cognitive development in children. The main causes of anaemia in children are malaria and inadequate intake of iron, folate, vitamin B12, or other nutrients. Other causes of anaemia include intestinal worms, haemoglobinopathy, and sickle cell disease. Although anaemia is not specific to malaria, trends in anaemia prevalence can reflect malaria morbidity, and they respond to changes in the coverage of malaria interventions (Korenromp 2004). Malaria interventions have been associated with a 60% reduction in the risk of anaemia when using a diagnostic cut-off of 8g/dl (RBM 2003).

Among eligible children age 6-59 months from interviewed households, most of them (94%) were tested for anaemia after consent from their parent or responsible adult (**Table 4.4**).

Table 4.5 shows 1% of children age 6-59 months are classified as having severe anaemia, defined as a haemoglobin concentration of less than 8 g/dl.

Patterns by background characteristics

- The prevalence of severe anaemia (haemoglobin <8g/dl) in children age 6-59 months is slightly higher in males as compared with females (1.6% versus 0.7%, respectively) (**Table 4.5**).
- Prevalence of severe anaemia in children aged 6-59 months is worse among households in the lower wealth quintiles compared to those in the higher wealth quintiles.
- Severe anaemia ranges from 0.8% in the Southern region to 1.5% in the Central region.

4.5 PREVALENCE OF MALARIA IN CHILDREN

Malaria prevalence in children

Percentage of children age 6-59 months classified as infected with malaria, according to microscopy results **Sample:** Children age 6-59 months

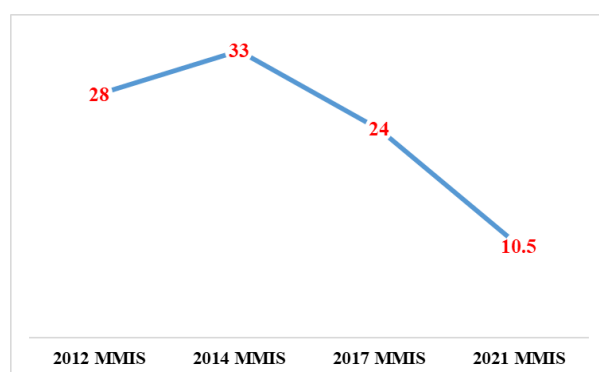
As is the case in many other countries in sub-Saharan Africa, malaria is the leading cause of death in Malawi among children under age 5. Malaria transmission is high throughout the year, contributing to development of partial immunity within the first 2 years of life. However, many people, including children, may have malaria parasites in their blood without showing any signs of infection. Such asymptomatic infection not only contributes to further transmission of malaria but also increases the risk of anaemia and other associated morbidity among the infected individuals.

In the 2021 MMIS, 10.5% of children age 6-59 months were positive for malaria parasites according to microscopy results (**Table 4.6**). Among children who tested positive, 86% had a *P. falciparum* infection (**Table 4.7**). Rapid diagnostic tests (RDTs) were done in conjunction with microscopy to facilitate the treatment of infected children during the survey fieldwork. Results from these RDTs are also presented in **Table 4.6** for reference. Twenty-four percent of children age 6-59 months tested positive for malaria antigens with RDTs. The differences in malaria prevalence observed between RDT and microscopy results are expected. Microscopic detection of malaria parasites depends on the visualization of the stained parasites under microscopy, whereas the diagnosis of malaria by RDT relies on the interaction between a parasite antigen present in the blood and an antibody in the RDT formulation.

Therefore, direct comparisons of malaria results from microscopy with those from RDTs should be avoided. The SD BIOLINE Malaria Ag P.f/Pan (HRP-II)TM rapid diagnostic test RDT, like many other commercially available RDTs, detects the *P. falciparum*-specific, histidine-rich protein-2 (HRP-2) rather than the parasite itself. The HRP-2 remains in the blood for up to a month after parasite clearance with antimalarials (Moody 2002). In areas highly endemic for *P. falciparum*, its persistence often leads to higher malaria prevalence estimates detected with RDTs compared with those measured by microscopy. Results presented in this section are based on the microscopy analysis of blood samples.

Figure 4.3 Trends in malaria prevalence among children

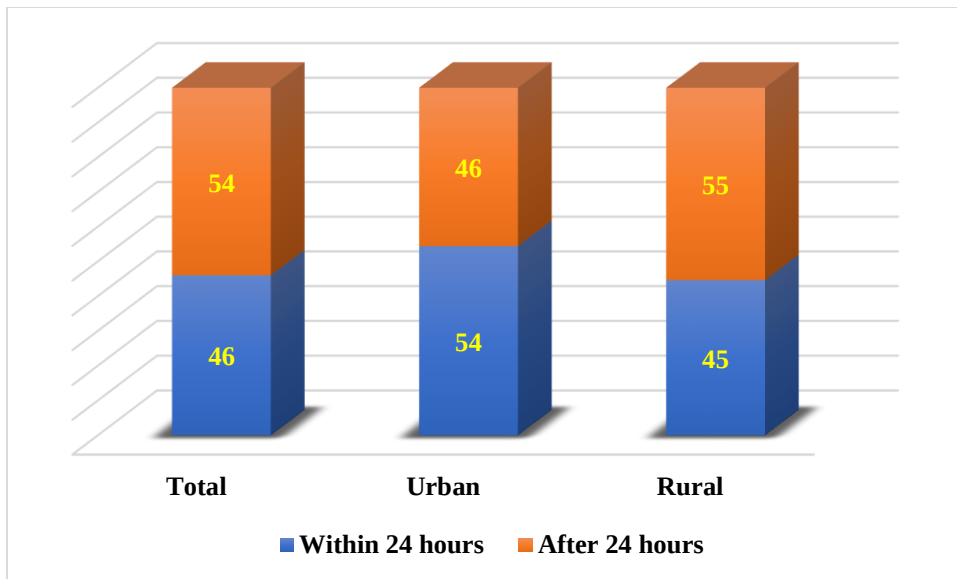
Percentage of children age 6-59 months who tested positive for malaria by Microscopy



4.6 MALARIA CARE SEEKING DURATION IN CHILDREN

Prompt diagnosis and timely treatment of malaria within 24 hours after onset of first symptoms can reduce illness progression to severe stages and therefore, decrease mortality. In 2021 MMIS, mothers and caregivers reported times taken for them to seek care for their children when they realize that they have malaria conditions. Figure 4.4 shows that nationally, 46% of under-five children were attended to within 24 hours of their the onset of first symptoms with majority (54%) of those in urban being attended within 24 hours compared to 45% in rural.

Figure 4.4 Malaria care seeking duration in children



Trends: National malaria prevalence in children under age 5 has decreased by 17.5 percentage points from 28% in 2014 to 10% in 2021. (Figure 4.3).

Figure 4.5 Prevalence of malaria in children by region

Percentage of children age 6-59 months who tested positive for malaria by microscopy

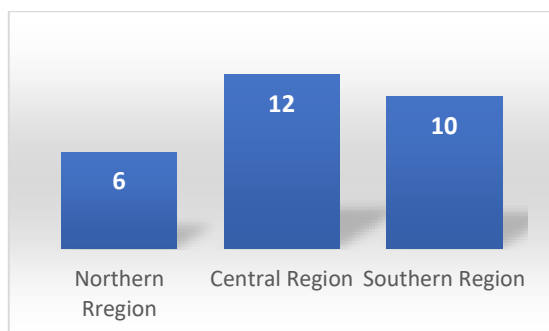
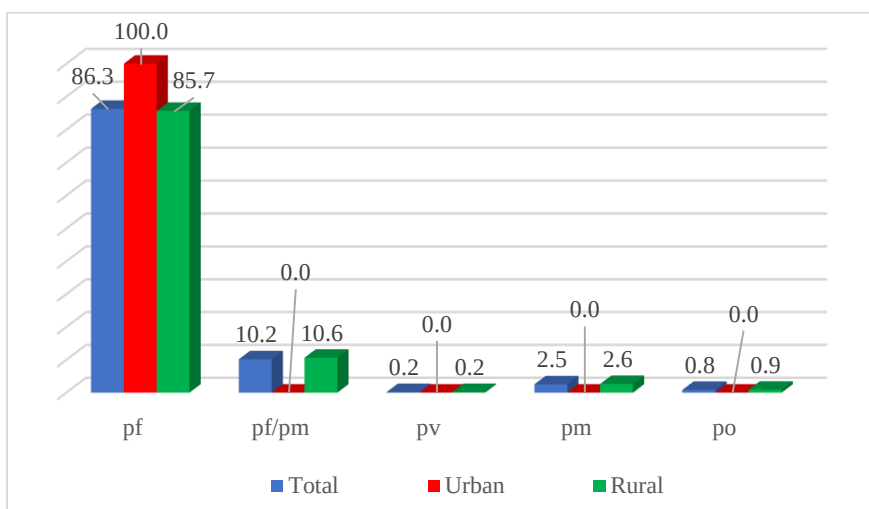


Figure 4.6 Distribution of malaria species in children

Percentage of children age 6-59 months who tested positive for malaria by malaria species



Eighty-six percent of underfive children had a Pf monoinfection, but 96.5% had a Pf infection (either mono or mixed species that included Pf) and 100% of those were in urban while 86% of them were in rural while other species contributed the remaining 4%.

Patterns by background characteristics

- Malaria prevalence by microscopy ranges from 4% in children age 9-11 months to 14% in children age 48-59 months (**Table 4.6**).
- Malaria prevalence by microscopy is higher among children whose mothers have no formal education (15%) than among those whose mothers have a secondary (4%) or higher (0%) education.
- Malaria prevalence by microscopy is about three times higher in rural areas (12%) than in urban areas (4%).
- By region, malaria prevalence by microscopy is lowest in the Northern region (6%) as compared with the Central (12%) and Southern (10%) regions. (**Figure 4.5**)
- The prevalence of malaria in children age 6-59 months by microscopy increases with decreasing wealth quintile, from 4% among those in the highest wealth quintile to 15% amongst those in the lowest wealth quintile.

LIST OF TABLES

For detailed information on malaria, see the following tables:

- **Table 4.1** Prevalence, diagnosis, and prompt treatment of children with fever
- **Table 4.2** Source of advice or treatment for children with fever
- **Table 4.3** Types of antimalarial drugs used
- **Table 4.4** Coverage of testing for haemoglobin and malaria in children
- **Table 4.5** Haemoglobin <8.0g/dl in children
- **Table 4.6** Prevalence of malaria in children
- **Table 4.7** Malaria species from microscopy

Table 4.1 Prevalence, diagnosis, and prompt treatment of children with fever

Percentage of children under age 5 with fever in the 2 weeks preceding the survey; and among children under age 5 with fever, percentage for whom advice or treatment was sought, percentage for whom advice or treatment was sought the same or next day following the onset of fever, and percentage who had blood taken from a finger or heel for testing, according to background characteristics, Malawi MIS 2021

Background characteristic	Children under age 5		Percentage for	Children under age 5 with fever	
	Percentage with fever in	Number of children		Percentage for whom advice or	Percentage who had blood

	the 2 weeks preceding the survey		whom advice or treatment was sought ¹	treatment was sought the same or next day ¹	taken from a finger or heel for testing	Number of children
Age in months						
<12	32.4	401	90.1	34.5	31.3	130
12-23	41.5	474	92.8	42.7	45.1	197
24-35	40.7	436	92.9	49.8	45.2	178
36-47	35.2	413	94.2	49.7	43.4	145
48-59	33.7	324	94.6	51.3	34.5	109
Sex						
Male	37.4	1,004	93.8	48.0	40.4	376
Female	36.7	1,044	92.1	43.1	41.5	383
Residence						
Urban	25.5	287	91.2	54.4	40.6	73
Rural	38.9	1,760	93.1	44.6	41.0	685
Region						
Northern Region	34.3	199	93.1	44.2	53.3	68
Central Region	37.6	860	91.2	36.8	36.2	323
Southern Region	37.2	988	94.4	53.5	42.8	367
Mother's education						
No education	35.2	226	89.2	22.1	33.0	79
Primary	41.0	1,338	92.6	46.3	43.6	549
Secondary	27.6	454	96.2	56.0	35.2	125
More than secondary	18.2	29	100.0	67.8	20.2	5
Wealth quintile						
Lowest	39.2	694	91.2	38.5	43.6	272
Second	40.4	588	92.2	43.0	42.0	238
Middle	36.9	376	95.3	52.4	37.5	139
Fourth	30.7	238	96.9	55.9	36.2	73
Highest	24.2	151	93.4	63.5	36.3	37
Total	37.1	2,047	92.9	45.5	40.9	759

¹ Includes advice or treatment from the following sources: [PUBLIC SECTOR, PRIVATE MEDICAL SECTOR, SHOP, MARKET, AND ITINERANT DRUG SELLER]. Excludes advice or treatment from a traditional practitioner

Table 4.2 Source of advice or treatment for children with fever

Percentage of children under age 5 with fever in the 2 weeks preceding the survey for whom advice or treatment was sought from specific sources; and among children under age 5 with fever in the 2 weeks preceding the survey for whom advice or treatment was sought, the percentage for whom advice or treatment was sought from specific sources, Malawi MIS 2021

Source	Percentage for whom advice or treatment was sought from each source:	
	Among children with fever	Among children with fever for whom advice or treatment was sought
Public sector		

Government hospital	210	210
Government health center	295	295
Government health post	49	49
Mobile Clinic	15	15
Fieldworker/CHW	40	40
Other public sector	0	0
Private medical sector		
Private hospital/clinic	15	15
Pharmacy	15	15
Private doctor	55	55
Mobile Clinic	3	3
Fieldworker/CHW	0	0
Other private medical sector	9	9
Other private sector		
Shop	0	0
Traditional practitioner	0	0
Market	0	0
Itinerant drug seller	0	0
Other	0	0
Number of children	705	705

CHW = Community health worker

Table 4.3 Type of antimalarial drugs used

Among children under age 5 with fever in the 2 weeks preceding the survey who took any antimalarial medication, percentage who took specific antimalarial drugs, according to background characteristics, Malawi MIS 2021

Background characteristic	Percentage of children who took:									Number of children with fever who took any antimalarial drug
	Any ACT	SP/ Fansidar	Chloro quine	Amodiaq uine	Quinin e pills	Quinine injection/I V	Artesuna te rectal	Artesuna te injection/I V	Other anti-malarial	
Age in months										
<6	75.0	75.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	42
6-11	92.6	92.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	65
12-23	84.0	84.0	0.0	0.0	0.0	0.0	0.0	0.8	0.0	155
24-35	94.7	94.7	0.0	0.0	0.0	0.0	0.0	1.3	0.4	153
36-47	92.1	92.1	0.0	0.4	0.0	0.0	0.0	0.0	1.0	126
48-59	88.7	88.7	0.7	0.0	0.0	0.0	0.0	0.0	1.5	83
Sex										
Male	89.5	89.5	0.2	0.1	0.0	0.0	0.0	0.6	0.0	319
Female	88.9	88.9	0.0	0.0	0.0	0.0	0.0	0.4	1.0	303
Residence										
Urban	70.3	70.3	1.0	0.0	0.0	0.0	0.0	0.0	0.9	64
Rural	91.4	91.4	0.0	0.1	0.0	0.0	0.0	0.6	0.4	558

Region										
Northern Region	84.4	84.4	0.0	0.8	0.0	0.0	0.0	0.0	0.0	56
Central Region	85.4	85.4	0.2	0.0	0.0	0.0	0.0	1.3	1.2	254
Southern Region	93.1	93.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	312
Mother's education										
No education	76.9	76.9	0.0	0.0	0.0	0.0	0.0	0.0	2.0	62
Primary	91.9	91.9	0.0	0.1	0.0	0.0	0.0	0.7	0.4	448
Secondary	85.7	85.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	110
More than secondary	70.2	70.2	17.1	0.0	0.0	0.0	0.0	0.0	0.0	4
Wealth quintile										
Lowest	88.7	88.7	0.0	0.2	0.0	0.0	0.0	0.6	1.1	222
Second	91.6	91.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	191
Middle	92.9	92.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	110
Fourth	83.1	83.1	0.0	0.0	0.0	0.0	0.0	3.0	0.0	67
Highest	79.0	79.0	1.8	0.0	0.0	0.0	0.0	0.0	1.7	34
Total	89.2	89.2	0.1	0.1	0.0	0.0	0.0	0.5	0.5	623

ACT = Artemisinin-based combination therapy

Table 4.4 Coverage of testing for anemia and malaria in children

Percentage of eligible children age 6-59 months who were tested for anemia and for malaria, by background characteristics (unweighted), Malawi MIS 2021

Background characteristic	Percentage tested for:			Number of children
	Anemia	Malaria with RDT	Malaria by microscopy	
Age in months				
6-8	89.2	89.2	89.2	83
9-11	93.1	93.1	93.1	87
12-17	95.5	95.5	95.5	222
18-23	97.5	97.5	97.5	202
24-35	93.5	93.5	93.5	429
36-47	93.2	93.2	93.2	427
48-59	95.2	95.2	95.2	333
Sex				
Male	94.2	94.2	94.2	898
Female	94.2	94.2	94.2	885
Mother's interview status				
Interviewed	94.3	94.3	94.3	1,602
Not interviewed, but in household	93.1	93.1	93.1	175
Not interviewed, and not in household ¹	100.0	100.0	100.0	6
Residence				
Urban	93.7	93.7	93.7	622
Rural	94.5	94.5	94.5	1,161
Region				
Northern Region				
Central Region				
Southern Region				

Mother's education²

No education	95.6	95.6	95.6	135
Primary	94.0	94.0	94.0	970
Secondary	95.0	95.0	95.0	462
More than secondary	88.6	88.6	88.6	35

Wealth quintile

Lowest	92.5	92.5	92.5	428
Second	94.8	94.8	94.8	423
Middle	94.4	94.4	94.4	356
Fourth	95.0	95.0	95.0	318
Highest	95.0	95.0	95.0	258

Total	94.2	94.2	94.2	1,783
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RDT = Rapid Diagnostic Test (SD Bioline Malaria Pf/Pan Ag)

¹ Includes children whose mothers are deceased

² For women who are not interviewed, information on education is taken from the Household Questionnaire. Excludes children whose mothers are not listed in the Household Questionnaire.

Table 4.5 Hemoglobin <8.0 g/dl in children

Percentage of children age 6-59 months with hemoglobin lower than 8.0 g/dl, according to background characteristics, Malawi MIS 2021

Background characteristic	Hemoglobin <8.0 g/dl	Num ber of childr en
Age in months		
6-8	1.8	94
9-11	1.0	90
12-17	2.6	227
18-23	1.6	206
24-35	0.7	416
36-47	1.0	404
48-59	0.4	347
Sex		
Male	1.6	881
Female	0.7	902
Mother's interview status		
Interviewed	1.2	1,629
Not interviewed, but in household	0.0	148
Not interviewed, and not in household	0.0	6
Residence		
Urban	0.3	241
Rural	1.3	1,543
Region		
Northern Region	1.3	185
Central Region	1.5	742
Southern Region	0.8	856
Mother's education²		
No education	1.1	191

	1.3	1,070
Primary		
Secondary	1.3	348
More than secondary	0.0	20

Wealth quintile

Lowest	1.2	593
Second	1.9	524
Middle	0.5	316
Fourth	0.5	215
Highest	0.1	135

	1.1	1,783
Total		

Note: Table is based on children who stayed in the household the night before the interview. Hemoglobin levels are adjusted for altitude using CDC formulas (CDC, 1998). Hemoglobin is measured in grams per deciliter (g/dl).

¹ Includes children whose mothers are deceased

² For women who are not interviewed, information on education is taken from the Household Questionnaire. Excludes children whose mothers are not listed in the Household Questionnaire.

Key Findings

- **Women's knowledge of malaria prevention method:** 92% of women age 15-49 reported that sleeping under a mosquito net can protect against malaria.
- **Exposure to malaria messages:** 1 or 2 in 10 women age 15-49 have seen or heard a malaria message in the past 6 months. Among those women, 51% saw or heard the message at a government clinic or hospital.
- **Mosquito net misuse:** Five percent of women age 15-49 reported nets being used for reasons other than sleeping

This chapter assesses the extent to which women age 15-49 are exposed to malaria messages and the channels through which they receive such messages. The chapter also provides data on women's basic knowledge of the causes, symptoms, and prevention of malaria. Net misuse is perceived to be a significant barrier to malaria prevention in Malawi and the 2021 MMIS asked direct questions on this topic. Information on the percentage of women who reported nets being used for purposes other than sleeping is included at the end of this chapter.

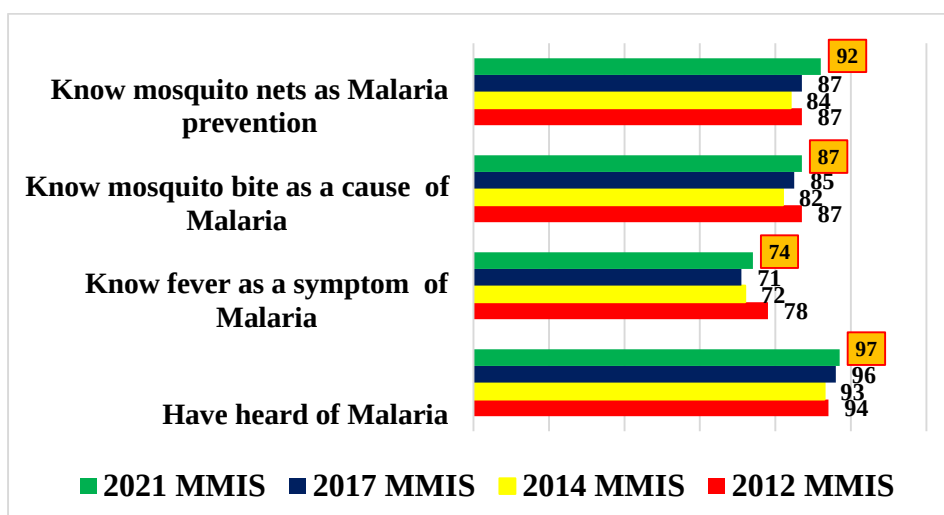
5.1 WOMEN'S KNOWLEDGE OF CAUSES, SYMPTOMS, AND PREVENTION

Perceptions, beliefs and attitudes about the causes of malaria, how to identify symptoms, and ways of preventing the illness are often overlooked in malaria control efforts. Yet such understanding is necessary to identify and target vulnerable populations and ensure the success of malaria control.

The majority of women age 15-49 in Malawi have heard of malaria (97%), recognize fever as a cause of malaria symptom (74%), report mosquito bites as a cause (87%), and say mosquito nets are a prevention method (92%) (Table 5.1).

Figure 5.1 Trends in malaria knowledge

Percentage of women age 15-49 who have heard of malaria and know malaria symptoms, causes, and prevention methods



Trends: In general, women’s knowledge about malaria, including causes, symptoms, and prevention, has not changed over the last 5 years. (Figure 5.1)

Patterns by background characteristics

- The percentage of women who report mosquito nets as a malaria prevention method ranges from 87% in the lowest wealth quintile to 96% in the highest wealth quintile (Table 5.1).
- Urban women (98%) are more likely to have heard of malaria than rural women (96%).

5.2 EXPOSURE TO MALARIA MESSAGES

Exposure to communication messages

Percentage of women age 15-49 who recall seeing or hearing a message about malaria through various sources in the past 6 months

Sample: Women age 15-49 who have seen or heard messages or information about malaria in the past 6 months

The effectiveness of social and behavioural change communications is notoriously difficult to measure. Because of this, the success of these initiatives is often measured by the percentage of the target population who recall hearing or seeing messages. Exposure is the critical first step in increasing knowledge of malaria prevention methods, as well as attitudes and practices about malaria. The target population’s ability to recall messages about malaria is an indicator of how widely communications have penetrated the target audience.

Overall, almost one-fifth of women age 15-49 in Malawi (17%) heard or saw a message on malaria in the 6 months prior to the survey (Table 5.2). The most common place to be exposed to these messages was at government clinics/hospitals; 5 in 10 women who saw or heard a message on malaria in the 6 months prior to the survey did so at a government clinic or hospital (51%). Radio is the second most common form of media for exposure to malaria messages (26%), while TV and newspapers/magazines account for just 3% and 1%, respectively (Table 5.3).

At most 10% of women age 15-49 who were exposed to malaria messages heard/saw messages related to “malaria is dangerous” and/or “sleeping under a mosquito net is important” (10% and 9%) (Table 5.4).

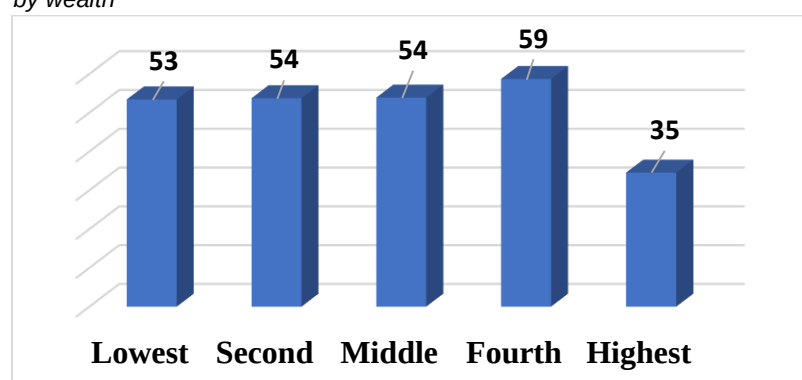
Trends: The percentage of women age 15-49 who have seen or heard malaria messages in the past 6 months increased from 22% in the 2014 MMIS to 31% in the 2017 MMIS, but declined to 17% in the 2021 MMIS. However, the average number of months ago that the messages were seen or heard also increased from 1.5 months in the 2017 MMIS to 1.8 months in the 2021 MMIS (**Table 5.2**).

Patterns by background characteristics

- Exposure to malaria messages at
 - government clinics/hospitals decreases as wealth quintile increases; from 53% in the lowest quintile to 35% in the highest quintile (**Figure 5.2**).
- Women living in urban areas (2%) were more likely to have heard or seen information about seeking treatment for fever than rural women (1%) (**Table 5.4**).
- Among women age 15-49 who saw or heard a message on malaria in the past 6 months, women from the Northern region (12%) were more likely than women in the Central (8%) and Southern (9%) regions to have heard or seen a message that “sleeping under a mosquito net is important” (**Table 5.4**).

Figure 5.2 Exposure to messages on malaria at government clinics/hospitals

Percentage of women age 15-49 who heard or saw a message on malaria at government clinics/hospitals by wealth



5.3 MOSQUITO NET MISUSE

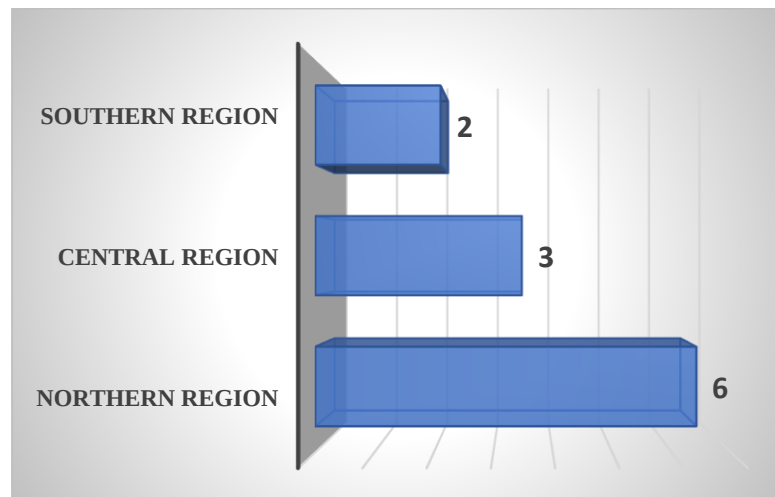
Use of an insecticide-treated net (ITN) is the primary strategy for vector control implemented by the NMCP in Malawi, and net misuse is considered a threat to Malawi’s Malaria Strategic Plan. However, only 5% of women age 15-49 reported using nets for reasons other than sleeping. Among women who reported mosquito nets in the house being used for reasons other than sleeping, almost four in ten women reported using nets for cover or protection (36%) (**Table 5.5**). Examples of using a net for cover or protection include use as fencing for vegetable gardens and/or chicken coops. One in twenty-five women reported their nets were also used for fishing (4%).

Patterns by background characteristics

- Women in rural areas (25%) were more likely to report nets being used as window screens than women in urban areas (16%).
- A higher percentage of women in the Northern (6%) region reported using a mosquito net for fishing than women in the Central and Southern region (4% and 2%, respectively) (**Figure 5.3**)

Figure 5.3 Map of women age 15-49 who reported using nets for fishing by region (NOTE THAT MAP WILL BE GENERAGED LATER FROM THE SAME FIGURE)

Among women who reported mosquito nets in the house being used for reasons other than sleeping, the percentage of women age 15-49 who reported mosquito nets being used for fishing by region



LIST OF TABLES

For detailed information on malaria, see the following tables:

- **Table 5.1 Women’s knowledge of malaria causes and symptoms**
- **Table 5.2 Messages about malaria**
- **Table 5.3 Exposure to messages on malaria by source**
- **Table 5.4 Exposure to specific messages and information on malaria**
- **Table 5.5 Use of mosquito nets for reasons other than sleeping**

Table 5.1 Women's knowledge of malaria causes, symptoms and prevention

Percentage of women age 15-49 who reported having heard of malaria, percentage who can recognize fever as a sign or symptom of malaria, percentage who reported mosquito bites as a cause of malaria, and percentage who reported that sleeping under a mosquito net can protect against malaria, according to background characteristics, Malawi MIS, 2021

Background characteristic		Percentage who have heard of malaria	Percentage who recognized fever as a symptom of malaria	Percentage who reported mosquito bites as a cause of malaria	Percentage who reported mosquito nets as a malaria prevention method	Number of women
Age	15-19	96.6	64.6	85.1	89.1	830
	20-24	97.1	77.2	86.9	92.8	781
	25-29	97.0	77.3	86.9	92.9	629
	30-34	97.6	78.5	87.5	93.7	458
	35-39	96.3	76.8	89.6	93.9	455
	40-44	97.0	74.7	89.6	91.4	324
	45-49	97.5	64.8	80.8	89.2	231
Residence	Urban	98.2	76.4	92.0	94.2	705
	Rural	96.7	72.8	85.5	91.3	3,004
Region	Northern Region	97.3	72.8	85.9	88.7	442
	Central Region	96.0	75.8	86.2	89.4	1,565
	Southern Region	97.8	71.7	87.5	95.0	1,702
Education	No education	90.8	66.3	72.8	83.4	355
	Primary	96.6	72.1	84.8	90.7	2,245
	Secondary	99.6	78.4	95.0	96.8	1,011
	Higher	100.0	82.9	98.7	99.5	99
Wealth quintile	Lowest	94.0	72.2	79.3	87.3	1,049
	Second	96.6	72.0	85.6	91.1	929
	Middle	98.8	72.8	89.3	94.3	756
	Fourth	99.2	78.5	94.2	95.3	527
	Highest	98.8	75.0	93.7	96.2	448
	Total	97.0	73.5	86.8	91.9	3,709

Table 5.2 Messages about malaria
Percentage of women age 15-49 who have seen or heard messages or information about malaria in the last 6 months, and among women who have heard or seen malaria messages in the last 6 months, percentage who have seen or heard malaria messages within the past 1 month, and average number of months ago the malaria message was seen or heard, according to background characteristics, Malawi MIS, 2021

Background characteristic		Percentage who have seen or heard malaria messages in the past 6 months	Number of women	Percentage who have seen or heard malaria messages in the past 6 months		
				Percentage who have seen or heard malaria messages in the past 1 month	Average months ago the messages were seen or heard	Number of women
Age	15-19	10.6	830	50.5	2.1	88
	20-24	16.8	781	53.0	1.8	131
	25-29	20.1	629	68.6	1.5	126
	30-34	20.0	458	48.0	2.2	92
	35-39	19.1	455	65.1	1.5	87
	40-44	25.5	324	51.3	1.8	83
	45-49	15.6	231	47.3	2.0	36
residence	Urban	19.0	705	56.0	1.9	134
	Rural	16.9	3,004	56.1	1.8	508
region	Northern Region	24.8	442	55.4	1.9	110
	Central Region	16.4	1,565	66.7	1.5	256
	Southern Region	16.3	1,702	46.6	2.0	277
Education	No education	12.6	355	51.0	2.0	45
	Primary	14.6	2,245	57.8	1.7	328
	Secondary	22.3	1,011	52.7	2.0	226
	Higher	44.8	99	66.2	1.5	44
Wealth quintile	Lowest	15.4	1,049	63.1	1.6	162
	Second	15.4	929	51.8	1.8	143

Middle	15.8	756	54.8	2.0	120
Fourth	20.1	527	50.2	2.0	106
Highest	24.9	448	58.6	1.7	112
Total	17.3	3,709	56.1	1.8	643

Table 5.3 Exposure to messages on malaria by source
Among women age 15-49 who saw or heard a message on malaria in past 6 months, percentage who saw or heard messages through various sources, according to background characteristics, Malawi MIS, 2021

Background characteristic		Exposure to malaria messages											N	w
		Government clinic/hospital	Community health worker	Friends/family	Work place	Drama groups	Peer educators	Poster/billboard	Television	Radio	Newspaper/magazine	Other		
Age	15-19	41.2	12.9	12.6	0.0	0.1	10.2	0.6	0.9	16.6	1.2	13.3		
	20-24	43.8	18.6	9.4	0.2	1.9	0.9	0.8	2.3	21.6	3.8	5.7		
	25-29	57.0	12.7	2.5	0.7	0.4	1.4	1.8	3.7	24.8	0.0	6.5		
	30-34	58.7	15.2	4.8	2.5	0.4	0.0	0.7	6.1	25.1	0.9	4.6		
	35-39	62.8	6.5	7.8	0.2	0.0	0.1	0.8	4.2	33.4	0.5	3.2		
	40-44	47.8	7.6	13.5	0.0	2.1	1.4	0.9	2.2	34.5	0.0	3.0		
	45-49	42.5	31.5	13.5	0.0	0.0	2.5	0.0	3.3	36.4	0.0	0.0		
residence	Urban	37.0	7.0	8.8	1.2	0.4	3.7	3.8	15.0	37.5	3.0	9.4		
	Rural	54.9	15.7	8.2	0.4	0.9	1.8	0.2	0.1	23.1	0.7	4.8		
region	Northern Region	52.9	15.8	7.6	0.5	0.6	5.4	1.6	3.8	21.5	0.0	5.5		
	Central Region	42.1	9.7	9.0	0.3	0.7	1.9	1.2	5.6	35.3	0.9	6.0		
	Southern Region	58.9	16.9	8.0	0.8	1.0	1.2	0.4	0.7	19.6	1.8	5.6		
Education	No education	50.1	15.6	16.8	0.0	0.0	1.1	0.4	0.0	23.4	0.0	1.8		
	Primary	52.1	18.0	8.6	0.0	0.2	1.7	0.3	0.4	21.8	0.4	6.4		
	Secondary	53.6	9.7	6.1	0.1	1.9	2.5	0.9	4.3	29.2	2.2	5.4		
	Higher	32.9	2.8	9.9	7.6	0.9	5.6	5.9	21.3	46.0	2.2	6.9		
Wealth quintile	Lowest	53.4	27.3	8.7	0.0	0.3	0.6	0.0	0.0	13.0	0.0	5.0		
	Second	53.7	10.0	8.5	0.0	1.7	0.6	0.0	0.0	25.6	0.3	4.5		

Middle	53.9	12.7	7.6	0.0	0.0	4.0	0.9	0.0	20.5	3.3	5.9	120
Fourth	58.7	6.3	10.1	1.8	1.7	2.3	1.8	0.9	33.4	0.5	7.2	106
Highest	34.5	7.8	6.9	1.5	0.4	4.5	2.7	17.6	45.1	2.1	6.9	112
Total	51.2	13.8	8.4	0.5	0.8	2.2	0.9	3.2	26.1	1.1	5.8	643

Table 5.4 Exposure to specific messages and information on malaria
Among women age 15-49 who saw or heard a message on malaria in the past 6 months, the percentage of women who heard or saw a specific message or information on malaria, according to background characteristics, Malawi MIS, 2021

		Exposure to malaria messages												Number of women
Background characteristic		Malaria is dangerous	Malaria can kill	Mosquito spread malaria	Sleeping under a mosquito net is important	Who should sleep under a mosquito net	Seek treatment for fever	Seek treatment for fever promptly (within 24 hours)	Importance of house spraying	Not plastering walls after spraying	Environmental sanitation activities	Other	Don't know	
Age	15-19	5.2	2.6	0.7	4.2	1.1	0.7	0.6	0.3	0.0	2.4	0.8	0.1	830
	20-24	7.7	3.0	1.8	9.9	0.3	0.8	1.4	0.4	0.5	3.3	1.5	0.7	781
	25-29	11.9	4.0	2.5	11.8	1.3	2.3	1.1	0.1	0.1	2.7	2.7	0.4	629
	30-34	10.3	5.1	1.2	12.7	2.5	0.8	4.6	0.0	0.0	2.8	0.9	0.2	458
	35-39	13.3	6.1	1.4	6.4	2.3	0.8	1.7	0.2	0.4	2.1	1.6	0.0	455
	40-44	14.9	6.9	3.5	13.9	2.0	4.3	3.6	0.2	0.8	3.4	0.6	0.4	324
	45-49	8.6	2.5	1.3	6.5	1.5	1.7	2.5	0.8	0.2	3.6	2.2	0.0	231
residence	Urban	10.4	5.2	2.4	11.1	2.1	1.9	2.3	0.5	0.2	2.7	2.1	0.1	705
	Rural	9.4	3.8	1.5	8.5	1.3	1.3	1.8	0.2	0.3	2.9	1.3	0.4	3,004
region	Northern Region	13.0	7.2	3.4	11.9	2.4	2.0	2.1	0.3	0.0	3.9	1.8	0.7	442
	Central Region	7.8	2.9	1.5	8.4	0.9	1.8	1.8	0.3	0.1	2.4	1.9	0.2	1,565
	Southern Region	10.3	4.3	1.4	8.7	1.6	0.8	1.9	0.3	0.5	3.0	0.9	0.3	1,702
Education	No education	7.0	3.6	0.7	4.5	0.0	1.0	0.5	0.0	1.2	0.1	0.7	0.7	355
	Primary	7.6	2.9	1.3	6.8	1.1	1.1	1.6	0.2	0.2	2.6	1.1	0.2	2,245
	Secondary	13.3	5.9	2.2	13.5	1.7	2.0	2.9	0.4	0.1	4.1	2.0	0.5	1,011
	Higher	25.1	13.0	8.3	29.2	9.1	3.7	3.6	1.5	0.5	5.0	5.9	0.0	99
Wealth quintile	Lowest	8.2	2.8	1.3	7.7	0.3	0.9	1.5	0.2	0.5	1.9	1.6	0.4	1,049
	Second	7.5	3.4	1.3	6.8	0.7	2.1	2.1	0.3	0.3	3.0	1.2	0.5	929
	Middle	9.4	3.6	0.8	8.9	1.9	0.9	1.2	0.0	0.0	2.3	1.1	0.1	756
	Fourth	12.8	4.9	2.7	10.8	2.5	1.0	1.9	0.4	0.1	5.0	1.1	0.2	527
	Highest	13.4	8.0	3.6	14.6	3.4	2.6	3.3	0.7	0.1	3.0	2.6	0.1	448
	Total	9.6	4.0	1.7	9.0	1.4	1.4	1.9	0.3	0.3	2.8	1.4	0.3	3,709

Table 5.5 Use of mosquito nets for reasons other than sleeping
Percentage of women age 15-49 who reported mosquito nets in the house being used for reasons other than sleeping, and what they were used for, by background characteristics, Malawi MIS, 2021

Background characteristic		Percentage who reported mosquito nets in the house being used for reasons other than sleeping	Number of women	Among those who reported mosquito nets in the house being used for reasons other than sleeping						Number of women who reported use of mosquito nets other than sleeping
				Fishing	Cover/ protection	Window screen	Clothing/ wedding veil	Other	Don't know	
Age	15-19	5.8	830	5.3	37.6	24.5	0.0	40.8	0.0	48
	20-24	3.2	781	7.0	31.9	29.9	0.0	31.1	0.0	25
	25-29	4.1	629	1.6	46.7	7.5	0.0	47.7	0.0	26
	30-34	3.5	458	0.0	33.9	50.7	0.0	23.1	0.0	16
	35-39	5.9	455	0.0	29.0	27.0	9.1	36.3	0.0	27
	40-44	3.1	324	12.8	23.0	9.1	1.3	59.9	0.0	10
	45-49	7.0	231	0.0	46.1	14.5	0.0	43.7	0.0	16
Residence	Urban	2.7	705	5.7	39.2	16.2	0.7	39.7	0.0	19
	Rural	5.0	3,004	3.3	36.0	24.6	1.6	39.5	0.0	149
Region	Northern Region	6.8	442	6.4	33.2	55.6	0.5	24.3	0.0	30
	Central Region	5.5	1,565	3.5	41.1	23.5	2.8	31.2	0.0	86
	Southern Region	3.0	1,702	2.1	30.3	5.3	0.0	62.3	0.0	52
Education	No education	5.2	355	9.2	9.5	11.8	0.0	71.7	0.0	19
	Primary	4.2	2,245	3.0	43.9	27.7	1.4	30.2	0.0	94
	Secondary	5.0	1,011	3.0	30.4	22.8	2.4	44.5	0.0	51
	Higher	4.8	99	0.0	57.6	0.0	0.0	42.4	0.0	5
Wealth quintile	Lowest	3.6	1,049	4.6	22.4	18.9	6.5	49.8	0.0	38
	Second	5.9	929	2.0	45.6	25.3	0.0	32.3	0.0	55
	Middle	3.7	756	6.1	33.0	28.4	0.0	39.4	0.0	28
	Fourth	4.8	527	4.3	37.4	31.9	0.5	30.8	0.0	25
	Highest	5.0	448	1.9	40.4	12.5	0.0	49.7	0.0	22
Total		4.5	3,709	3.6	36.4	23.7	1.5	39.5	0.0	168

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